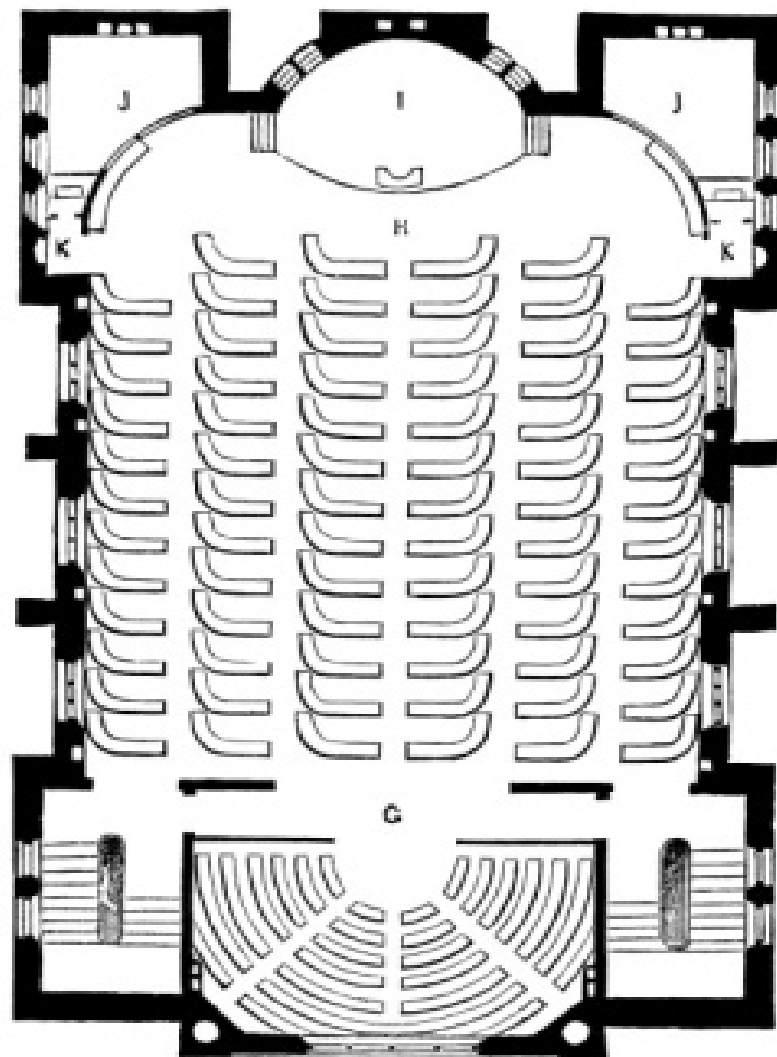
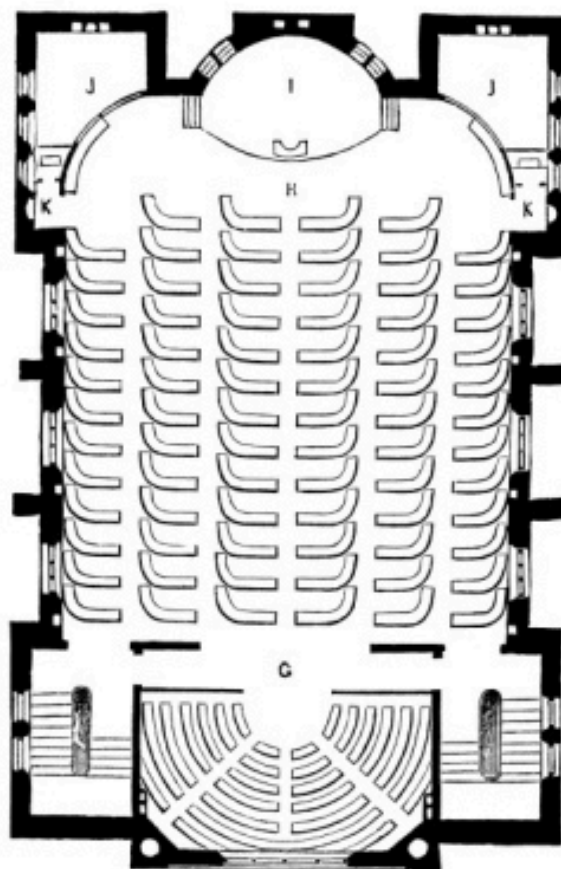


THE
ARCHITECTURAL REVIEW
AND
AMERICAN
BUILDERS' JOURNAL.



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Title: The Architectural Review and American Builders' Journal, Aug. 1869

Author: Various

Editor: Samuel Sloan

Release date: December 22, 2019 [eBook #60997]

Most recently updated: October 17, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/60997

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*** START OF THE PROJECT GUTENBERG EBOOK THE
ARCHITECTURAL REVIEW AND AMERICAN BUILDERS'
JOURNAL, AUG. 1869 ***

**THE
ARCHITECTURAL REVIEW
AND
AMERICAN
BUILDERS' JOURNAL.**

VOL. II.—Entered according to Act of Congress, in the year 1869, by
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MONTHLY REVIEW.

THE LONDON BUILDER AND OURSELVES.

In a tolerantly critical notice of the REVIEW recently published in the *Builder*, we find an effort to substantiate a charge formerly made by it, and replied to by us, on the subject of “trickery” in the construction of the exteriors of American buildings. The *Builder* reiterates the charge and points to Grace Church, New York, in proof of the truth of it. That marble edifice, he avers, has a wooden spire, crocketed, etc., painted in imitation of the material of which the body of the church is constructed. Alas, we must acknowledge the wood. And we will make a clean breast of it, and still farther acknowledge that at the time that Grace Church was built, our land of wooden nutmegs, and other notions, had not an architectural idea beyond the wooden spire, and that our city and country churches, that aspired at all, were forced to do so in the national material of the day. That said sundry spires of wood were *of necessity*, painted, is most true; and furthermore, white-lead being a great favorite with the people generally, [when our manners, customs, and tastes were more immaculate than in these degenerate days of many colors,] that pigment was the ruling fashion. That the color of the marble, of which Grace Church’s body is constructed, should be similar to that with which said ecclesiastical edifice’s spire was coated, is unfortunate; but, that the resemblance goes to prove any attempt at a *cheat*, we most strenuously deny. Grace Church is of a by-gone taste,—an architectural era which we now look back to in order to see, by contrast, how far we have advanced in architectural construction. Trinity Church, New York, was the first great effort at a stone spire which our Architects ventured to rear. And although hundreds have followed its lead, none in this soaring republic have gone so near to heaven as that yet. But the thing once effected is sure to be improved upon.

We are not at all abashed then, to own to the *wooden spire painted to imitate stone*, which crowns the steeple of old Grace Church, New York. And the less annoyance should it give our most sensitive feelings, when we reflect that the dome of the great ST. PAUL’S, London, is no less a delusion

and a cheat, it being of wood, coated with lead and painted on the outside, having a false dome on the inside, considerably smaller than the external diameter would naturally lead the confiding observer to expect. The body of St. Paul's is of stone. Why, according to the requirements of the *Builder*, is not the dome, like that of the Pantheon at Rome, likewise of stone?

Do we suppose, for an instant, that Sir Christopher Wren was guilty of a deliberate cheat in so constructing it? Certainly not. He used the material which he considered best suited to his purpose and his means. And so we should, in charity, suppose did the Architect of Grace Church, New York.

The *Builder*, like too many of our English cousins, who do us the honor of a visit, falls into error in supposing that wood is generally used for ornamentation of exteriors. In none of our larger cities is this the case. And when that critical and usually correct authority says, "Even the Fifth avenue itself is a sham as to much of its seeming stone-work," it displays a melancholy absence of its uniform discernment, judgment, and sense.

The only other constructive material to be found on the fronts of the Fifth Avenue, New York, besides marble, brown stone, or pressed (Philadelphia) brick, is in the gutter, which is either of zinc or galvanized iron, and forms the upper portion of the cornice.

Porches and Hall-door frontisces, of every style, are of marble or stone, and never of wood. Pediments and all trimmings around windows are invariably of stone. In fact we are not a little surprised at the apparent want of information on this subject by so well posted an observer as the *Builder* is acknowledged to be. Some twenty years ago the taunt might lie most truthfully applied to our efforts at architectural construction, but to-day the "trick" of painted and sanded wood would be hissed down by our citizens who claim to live in residences the majority of which are greatly superior to residences of the same class in London, as far at least as material is concerned. No, no—criticism to be useful must be just; and to be just must be founded strictly on truth unbiassed by prejudice.

We do not desire in these remarks to throw the slightest doubt on the good intentions of the London *Builder* in its monitorial check, but our wish is to correct the erroneous information which it has received, and which has led to the mistake under which it evidently labors.

We as utterly despise any falsehood in construction as our honestly outspoken contemporary, and will at every opportunity disclose and denounce its adoption in this country in all cases where there is any pretension to architectural design. For a new country like this, it is at least creditable that, even in a small class of dwellings, the architect is, as a general thing, called on to design and frequently to superintend—every thing is not left to the builder as in London. Yet there is and always will be in this as in all other countries a large class of private buildings outside the pale of legitimate taste; creations ungoverned and ungovernable by rule. But such should never be taken as examples of the existing state of the constructive art of the day; they should rather prove the unfortunate exceptions to the fact of its position. Even these it will be our duty to watch over and try to set right; for we are ardent believers in the influential power of information, and look with assurance to the education of our people generally on this subject of judgment and taste in building as the infallible means of turning to good account the remarkable progress in that constructive art of the American nation, which the observant London *Builder* notices with the generous well-wishing of a kindly professional brother.

THE MANSARD MADNESS.

Of all the intellectual qualifications which man is gifted with, there is not one as sensitive as that which enables him to discern between what is intrinsically good, and what is bad or indifferent to his eye. Yet are there none of all man's mental attributes so frequently and so grossly outraged as is this to which we now allude, called Taste.

Custom has much to say in the question of arbitrary rule which taste so imperatively claims. Persistence in any thing will, of necessity, make itself felt and recognized, no matter how odious at first may be the object put before the public eye, and ultimately that object becomes what is commonly called "fashionable." This apparent unity of the public on one object is variable and will soon change to another, which in its turn will seem to reign by unanimous consent and so on *ad infinitum*.

In Architecture this fickle goddess, Fashion, seems to reign as imperatively and as coquettishly as in any or all the affairs of this world of

humanity. That which was at first esteemed grotesque and ridiculous, becomes in time tolerable and at last admirable. But the apathy which sameness begets cannot long be borne by the novelty worshippers, and accordingly new forms and shapes remodel the idea of the day, until it ceases to bear a vestige of its first appearance and becomes quite another thing.

Of all the prominent features of architecture that which has been least changeable until late years is the "roof." The outline of that covering has been limited to a very few ideas, some of which resolved themselves into arbitrary rules of government from which the hardiest adventurer was loath to attempt escape.

Deviating from the very general style of roof which on the section presents a triangle, sometimes of one pitch, sometimes of another, but almost universally of a fourth of the span, the *truncated* form was to be found, but so exceedingly sombre was this peculiar roof that it never obtained to any great extent, and indeed it presented on the exterior a very serious obstacle to its adoption by architects in the difficulty of blending it with any design in which spirit, life, or elegance, was a requisite.

There are occasionally to be found in Europe, and even in America, examples of these truncated roofs, but it is very questionable whether there are to be met with any admirers of their effect.

The principle on which they are constructed has, however, a very great advantage in the acquirement of head-room in the attics, giving an actual story or story and half to the height, without increasing the elevation of the walls. The architects of the middle ages took a hint from this evident advantage, and used the truncated roof on their largest constructions. Its form is that of a pyramid with the upper portion cut off (*trunco*, to cut off, being its derivation.)

MANSART, or as he is more commonly called MANSARD, an erratic but ingenious French architect, in the seventeenth century invented the curb roof, so decided an improvement on the truncated that it became known by his name. This roof adorning the palatial edifices of France soon assumed so much decorative beauty in its curb moulding and base cornice, as well as in the dormers and eyelets with which it was so judiciously pierced, that it became a source of artistic fascination in those days in France; and as

Germany was indebted to French architects for her most prominent designs, the Mansard roof found its way there, and into some other parts of Europe.

But, much as English architects admired, as a whole, any or all of those superb erections of the Gallic Capital, it was a century and a half before it occurred to them to imitate them even in this most desirable roof.

Our architects having increased with the demand for finer houses and more showy public buildings, and having parted company with their Greek and Roman idols to which their predecessors had been so long and so faithfully wedded, and acknowledging the necessity for novelty, ardently embraced the newly arising fashion and the Mansard roof arose at every corner in all its glory. At first the compositions which were adorned with this crowning were pleasing to the general view, if not altogether amenable to the strict rules of critical taste. But in due time (and alas that time too surely and severely came) the *pseudo* French style with its perverted Mansard roof palled upon the public taste for the eccentricities its capricious foster-fathers in their innate stultishness compelled it to display.

Some put a Mansard roof upon an Italian building, some on a Norman, and many, oh, how many, on a Romanesque! Some put it on one story erections and made it higher than the walls that held it, in the same proportion that a high crowned hat would hold to a dwarf. Some stuck on towers at the corners of their edifices and terminated them with *Mansard domes*! Some had them inclined to one angle, some to another; some curved them inward, some outward, whilst others went the straight ticket.

The dormers too came in for a large share of the thickening fancies and assumed every style or no style at all. The chimney shafts were not neglected. Photos of the Thuilleries were freely bought up, and bits and scraps of D’Lorme were hooked in, to make up an original idea worthy of these smoky towers. “Every dog will have his day,” is a fine old sensible remark of some long-headed lover of the canine species, and applies alike to animals, men, and things. That it particularly applies to that much abused thing called the Mansard roof is certain, as the very name is now more appropriately *the absurd roof*.

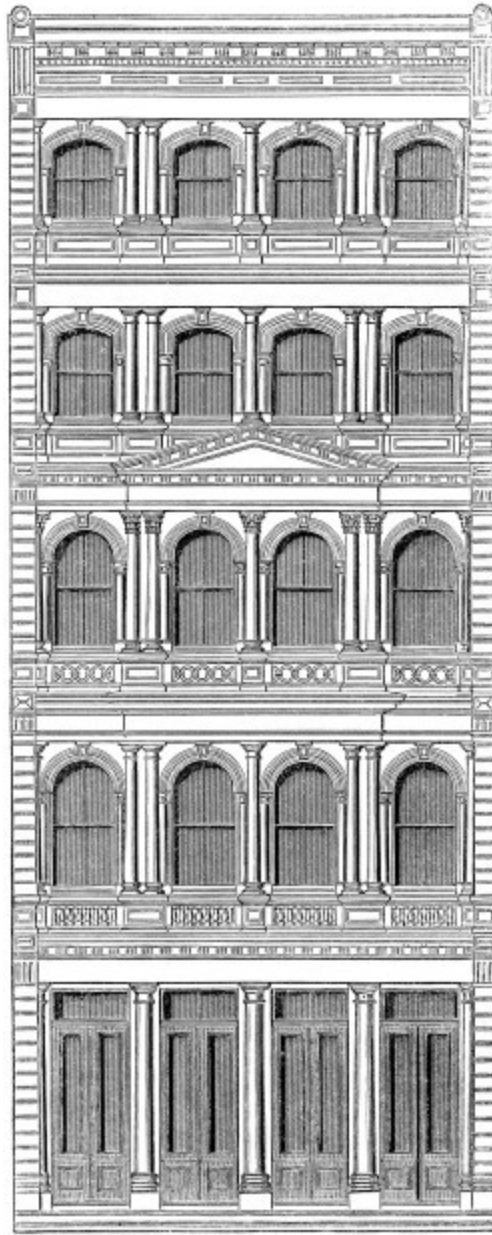
Fashion begins to look coldly upon her recent favorite, which in truth “has been made to play such fantastic tricks before high Heaven, as make the angles weep;” and it is doomed.

A few years hence, and we will all look back in amused wonder at the creations of to-day, crowned with the tortured conception of MANSARD.

HYDRAULIC CEMENT.

The rapid hardening under water of the cement which from that property derives its name of "Hydraulic Cement," has been, and indeed is still, a subject of discussion as to the true theory of such action. We find in the June number of the *Chemical News* a paragraph which must prove very interesting to manufacturers as well as to all who use and take an interest in that most useful of building materials to which the Architect and the Engineer are so deeply indebted.

"In order to test the truth of the different hypotheses made concerning this subject, A. Schulatschenko, seeing the impossibility of separating, from a mixture of silicates, each special combination thereof, repeated Fuch's experiment, by separating the silica from 100 parts of pure soluble silicate of potassa, and, after mixing it with fifty parts of lime, and placing the mass under water, when it hardened rapidly. A similar mixture was submitted to a very high temperature, and in this case, also, a cement was made. As a third experiment, a similar mixture was heated till it was fused; after having been cooled and pulverized, the fused mass did not harden any more under water. Hence it follows that hardening does take place in cement made by the wet as well as dry process, and that the so-called over-burned cement is inactive, in consequence of its particles having suffered a physical change."



IRON STORE-FRONTS, No. V.

By WM. J. FRYER, JR., WITH
MESSRS. J. J. JACKSON & BROS., NEW YORK.

NATIVE COLORED MARBLES.

In the preceding number we have spoken in general terms of this beautiful acquisition to our art materials, and indeed we feel that we cannot esteem this new American discovery too highly; for even in Europe such stone is extremely scarce at the present day, and it is fortunate that the location in which the quarries exist is open to the Old World to freely supply the wants of its artists, as well as our own. The beautiful Lake Champlain affords excellent commercial facilities, the Chambly Canal and Sorel River improvements opening a free navigation both with the great chain of lakes, and the Atlantic Ocean. The Champlain Canal connecting it with the Erie Canal and Hudson River, giving it uninterrupted communication with New York State and its Empire City, from the latter end of March to the middle of December.

The quarry is situated in a great lode projecting up in the bosom or bay of Lake Champlain, forming an island of several acres outcropping on each shore, and giving evidence that the deposit extends and really forms, at this point, the bed of the lake, its supply being thought to be inexhaustible.

The marble occurs in beds and strata varying in thickness from one to six feet, and will split across the bed or grain; blocks of any required size being readily obtained. Its closeness of texture and hardness render it susceptible of a very high polish, and it will resist in a remarkable degree all atmospheric changes. It is hard to deface with acids or scratches, and this one fact should attach to it much additional value. Its variegation in color, as shown by the specimens taken from its outcroppings, give promise of a much richer development as the bed of the quarry is approached; and must equal in beauty and durability the highly prized oriental marble of ancient and modern times.

The facilities, already alluded to, of its transportation to all the markets for such material in the country and to the seaboard, whence it can be shipped to any part of the world, must tend to bring it into general use here and elsewhere, that colored marbles are required for building and ornamental purposes.

We are much indebted to a gentleman of Philadelphia, whose taste and liberal enterprise have so opportunely brought to our knowledge this most remarkable deposit of one of Nature's most beautiful hidden treasures,

which must, at no distant day, add vastly and more cheaply to the art material of our country.

THE palace in course of construction at Ismalia, for the reception of the Empress Eugenie during her stay in Egypt, will be 180 feet wide and 120 deep. The estimate cost is 700,000fr. According to the contract it is to be finished by the 1st of October, for every day's delay the architect will be subject to a fine of 300fr per day, and if finished before he will receive a bonus of 300fr per day. The building will be square; in the centre there is to be a dome covered with Persian blinds. On the ground floor there will be the ball, reception, and refreshment rooms. An idea can be formed of the importance of this structure and of the work necessary to complete it within the required time, as it will contain no less than 17,400 cubic feet of masonry.

TO REMOVE WRITING INK—To remove writing ink from paper, without scratching—apply with a camel's hair brush pencil a solution of two drachms of muriate of tin in four drachms of water; after the writing has disappeared, pass the paper through the water and dry.

DESCRIPTIONS.

IRON STORE FRONTS, No. 5.

BY W. J. FRYER, JR., NEW YORK.

The elevation, shown in the [accompanying page illustration](#), shows an iron front of five stories, having a pedimented centre frontispiece of three stories in *alto relievo*.

The style, though not in strict accordance with rule, is showy, without being objectionably so, and goes far to prove the capabilities of iron as a desirable material in commercial Architecture, where strength, display, and economy may be very well combined.

Such an elevation as this, now under consideration, could not be executed in cut stone, so as to produce the same appearance, without incurring a much greater expense, and in the event of a continuous block of such fronts, the balance of economy would be wonderfully in favor of the iron, for the moulds could be duplicated and triplicated with ease, whilst the same composition executed to a like extent in stone would not be a cent cheaper in proportion. Every capital and every truss, and every fillet, should be cut in stone independently of each other, no matter how many were called for.

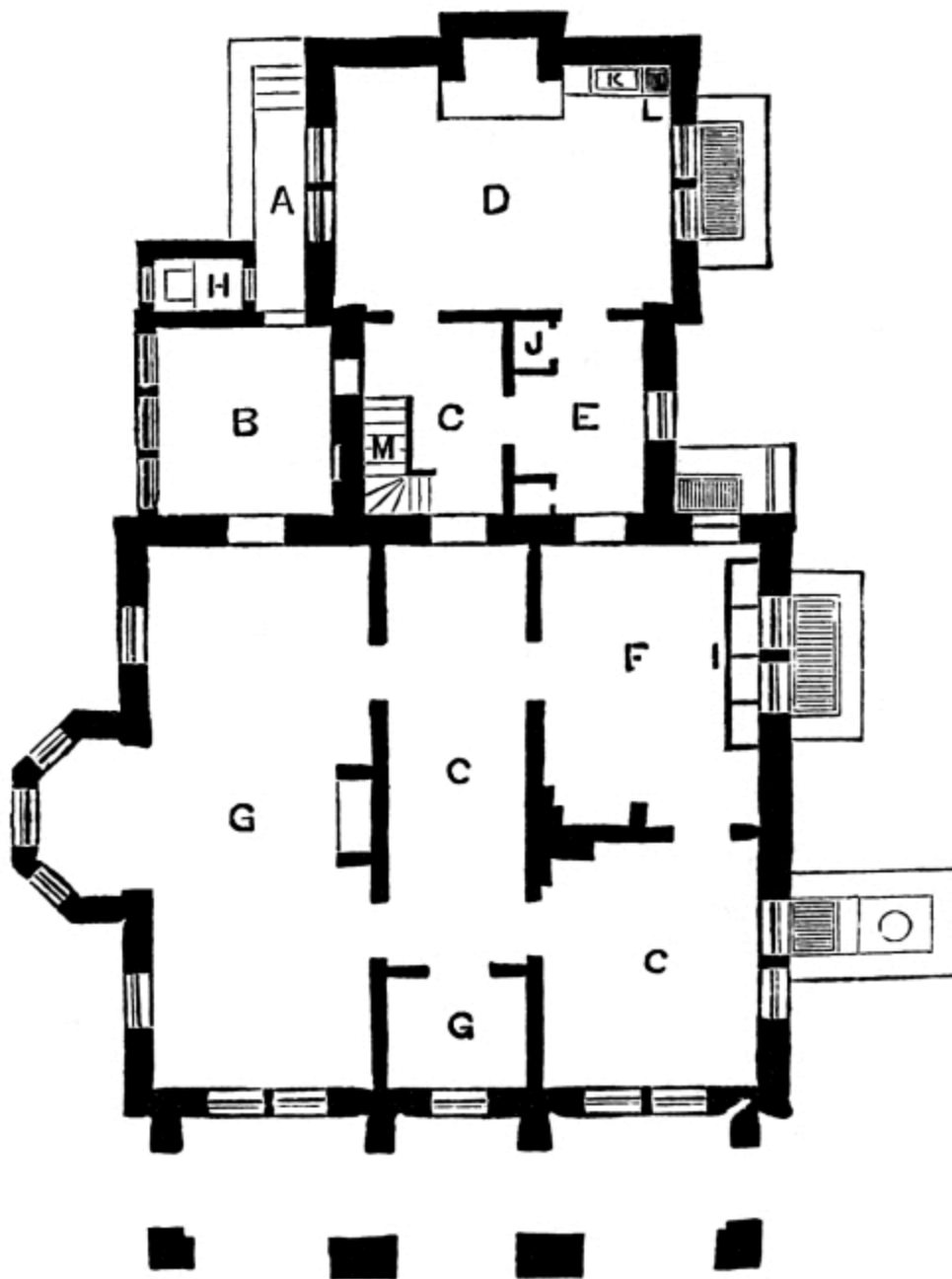
It may be very well to say that stone is the proper material, according to the long-accepted notion of art judgment, and that iron has to be painted to give it even the semblance of that material, being, therefore, but a base imitation at best. All very true. But, nevertheless, iron, even as a painted substitute, possesses advantages over the original material of which it is a copy, rendering it a very acceptable medium in the constructive line, and one which will be sought after by a large class of the community who desire to have this cheap yet practical material, even though it be not that which it represents. As a representative it is in most respects the peer of stone though not it identically.

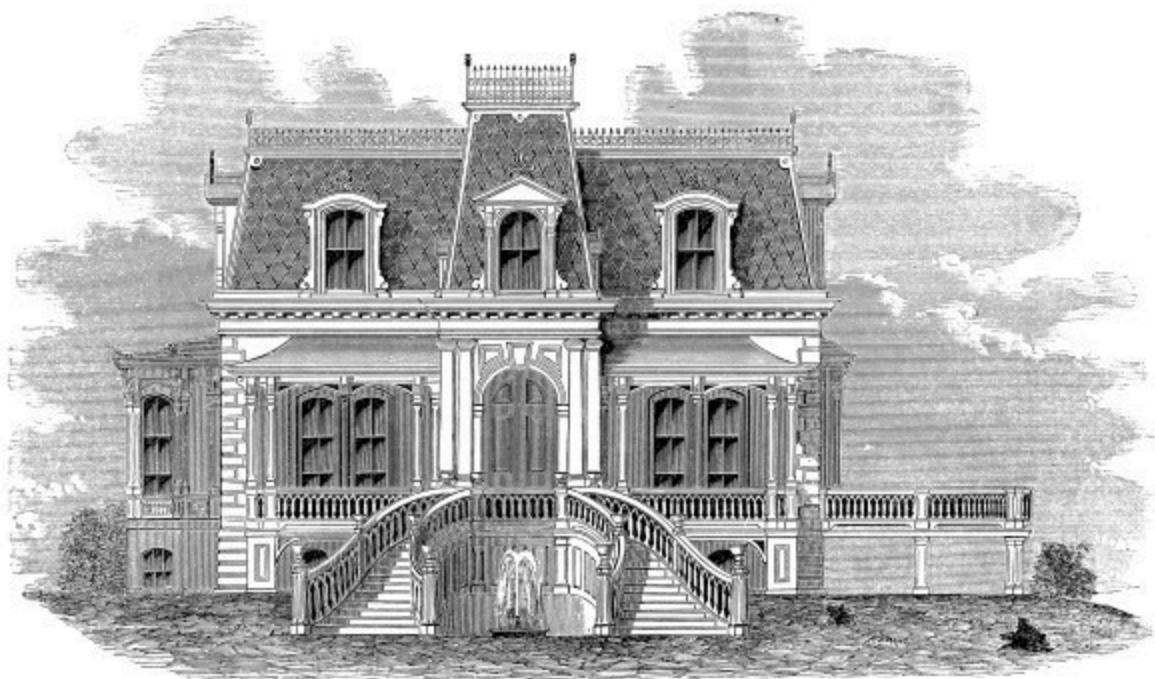
SUBURBAN RESIDENCE IN THE FRENCH STYLE.

BY CARL PFEIFFER, ESQ., ARCHITECT, N. Y.

This design is of one of those homes of moderate luxury wherein the prosperous man of business may enjoy in reason the fruits of his energetic toil. There is nothing about it to indicate presumptuous display, but rather the contented elegance of a mind at ease, surrounded with unostentatious comfort.

FIG. 1.





SUBURBAN RESIDENCE IN THE FRENCH STYLE.

CARL PFEIFFER, ESQ., ARCHITECT, NEW YORK.

On the westerly slope of the Palisades, and two miles to the west of the Hudson, [this residence](#) was built by one of New York's retired merchants.

It is sixteen miles from Jersey City, in a town of but a few years growth, named "Terraflly," in Bergen county, and stands on a hill commanding some of the most charming pieces of pastoral scenery, occupying about thirty acres laid out in lawns, walks, gardens, etc., and tastefully ornamented with shrubbery, having a fountain on the lawn in front of the house (as shown.)

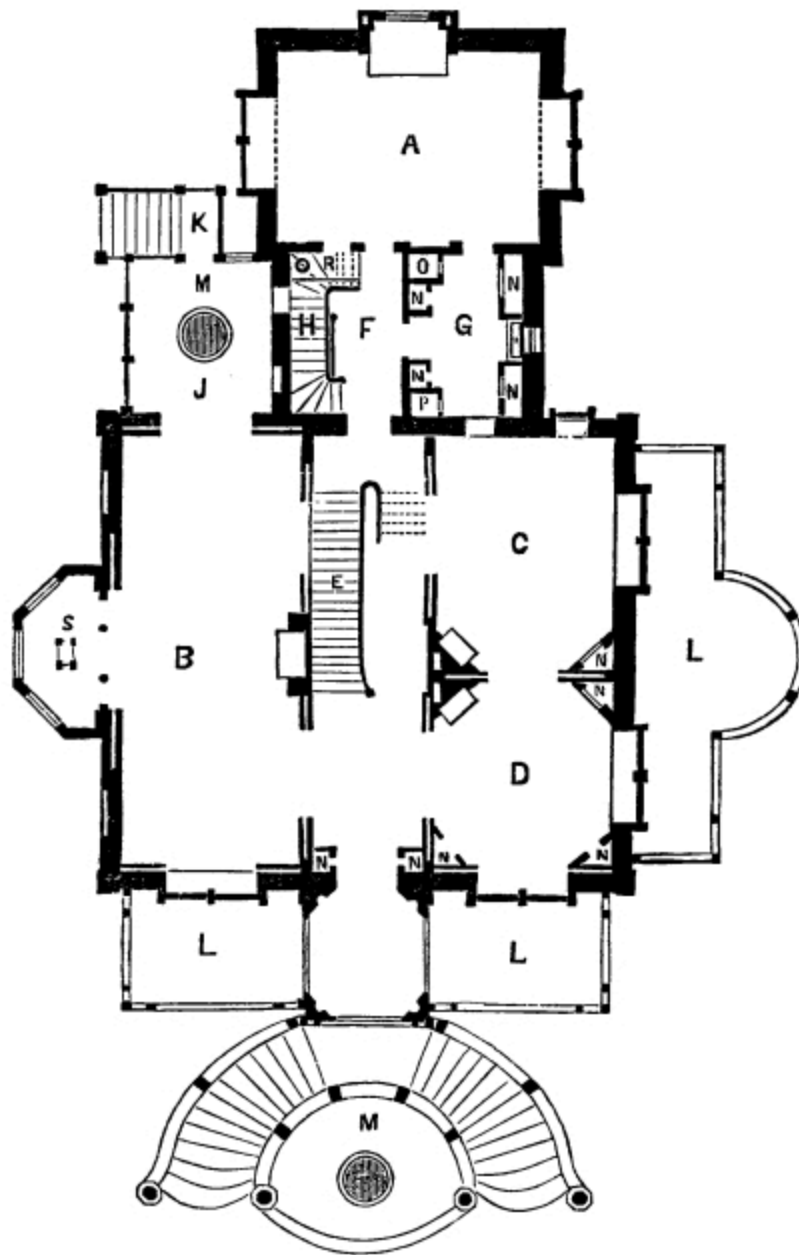
The approach is from the public road, by a drive through a grove of about ten acres of stately trees, passing by the side of a pretty pond formed by the contributions of several streams and making a considerable sheet of water. About the middle of this pond the sides approach so near to each other as to be spanned by an artistic little stone arched bridge which leads to the garden.

From the house one looks on a lovely panorama of inland scenery. The Palisades towards the east, the Ramapo mountains to the northwest; and

looking in a southerly direction the numerous suburban villages and elegant villas near New York may be seen.

The [house](#) is constructed of best Philadelphia pressed brick with water-table, quoins, and general trimmings of native brown stone neatly cut. It stands high on a basement of native quarry building stone and has for its foundation a permanent bed of concrete which likewise forms the basement floors, as well as a durable bedding for the blue flagging of Kitchen and Laundry hearths.

FIG. 2.

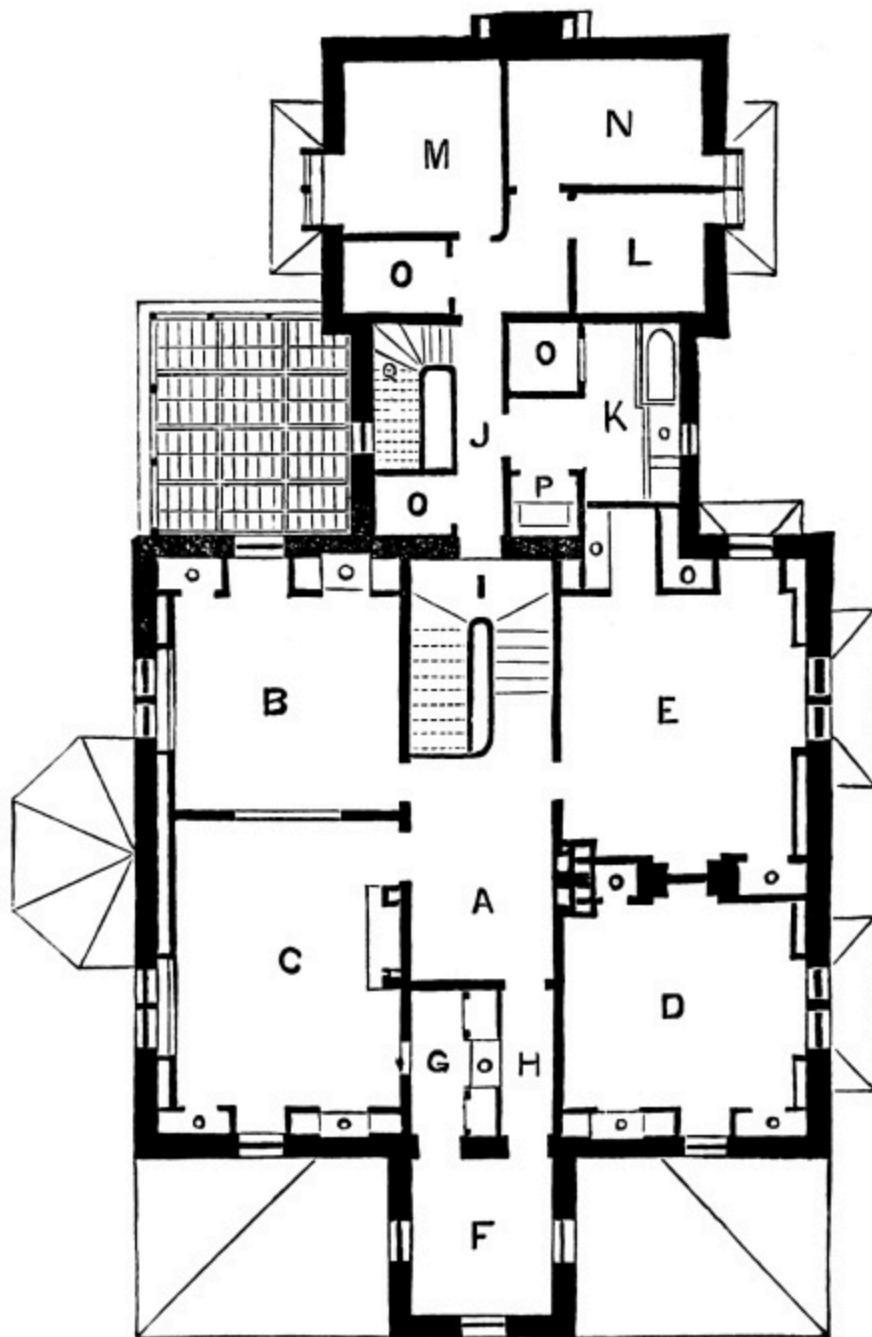


The arrangement of plan is admirably calculated to conduce to the comfort of the family. It is as follows:

[Fig. 1](#) shows the plan of the basement. A, steps and passage leading from Yard. B, Servant's Dining Room. C, C, C, Coal Cellar and Passages. D, Kitchen. E, Pantry. F, Laundry. G, G, Cellars. H, Water Closet. I, Wash tubs in Laundry. J, Dumb waiter. K, Wash-tray. L, Sink. M, Back stairs.

[Fig. 2](#) shows the plan of the principal story. A, Dining Room. B, Drawing Room. C, and D, Parlors connected by sliding doors with the Drawing Room through the hall. E, Principal staircase. F, Back Hall. G, Butler's Pantry with dumb waiter, plate closet, wash-trays, etc. H, Back stairs. J, Conservatory. K, Steps leading down to Yard. L, L, L, Verandahs. M, M, Piscinæ.

FIG. 3.



[Fig. 3](#) shows the arrangement of the Chamber floor, or second story. A, the Hall. B, C, D, and E, Chambers. F, Boudoir. G, Closet. H, Passage to Boudoir. I, Half landing connected with rear addition. J, Back passage. K, Bath Room. L, M, N, Servant's Bed Rooms. O, O, O, Clothes Closets. P, Water Closet, o, o, o, o, o, o, Wardrobes in the several Chambers. These

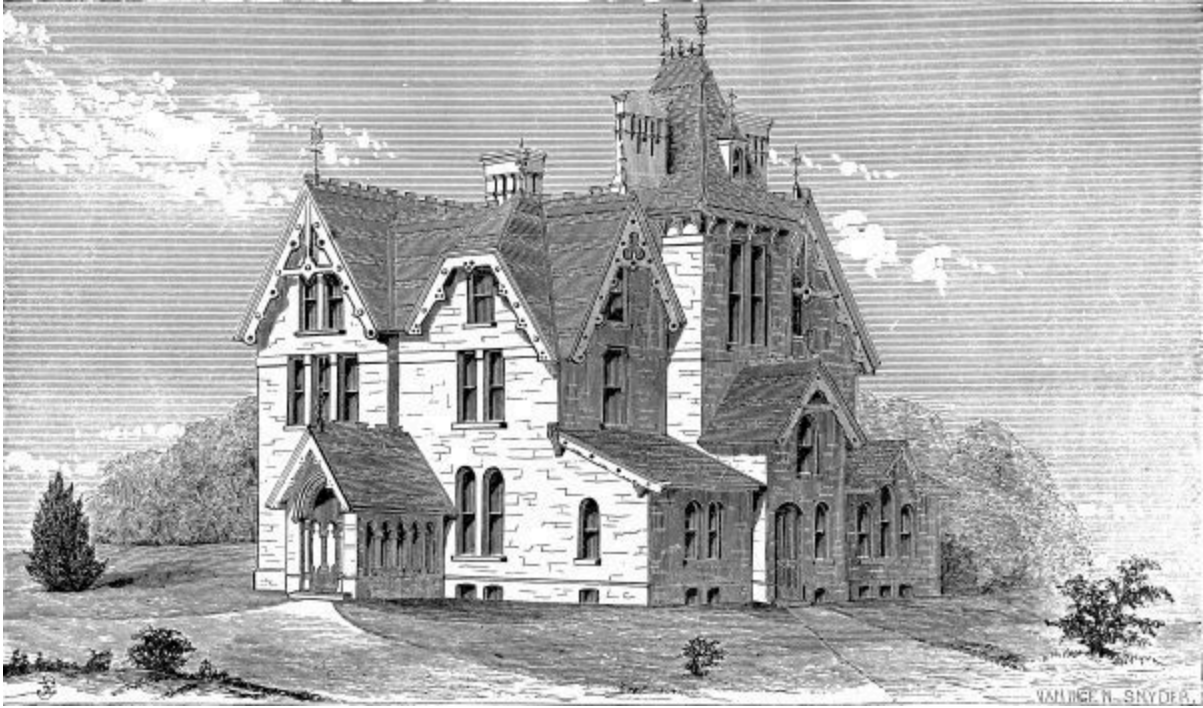
occupy the angle enclosed by the slope of the Mansard, thus leaving the walls of the chambers plumb.

The roof is flat, and is embellished at the curb with a rich traceried iron balustrade, making a safe and desirable promenade platform. All the accessories that go to make a comfortable home are provided, and the whole forms a model retreat from busy life to Nature and her charms.

SUBURBAN RESIDENCE IN THE FRANCO-GOTHIC STYLE.

We here give a perspective view of a [capacious suburban residence](#), showing the marked effects of light and shade produced by means of Gothic gables on a building of a square plan. A hipped roof on such a plain form would make a most uninteresting mass of heaviness. The judicious addition of bay windows is always desirable in such compositions; and the hooded gables give a pleasing quaintness to the whole. We present, on next page the principal floor plan, which is somewhat unusual in arrangement, but comfortable, as such form of house is always sure to be.

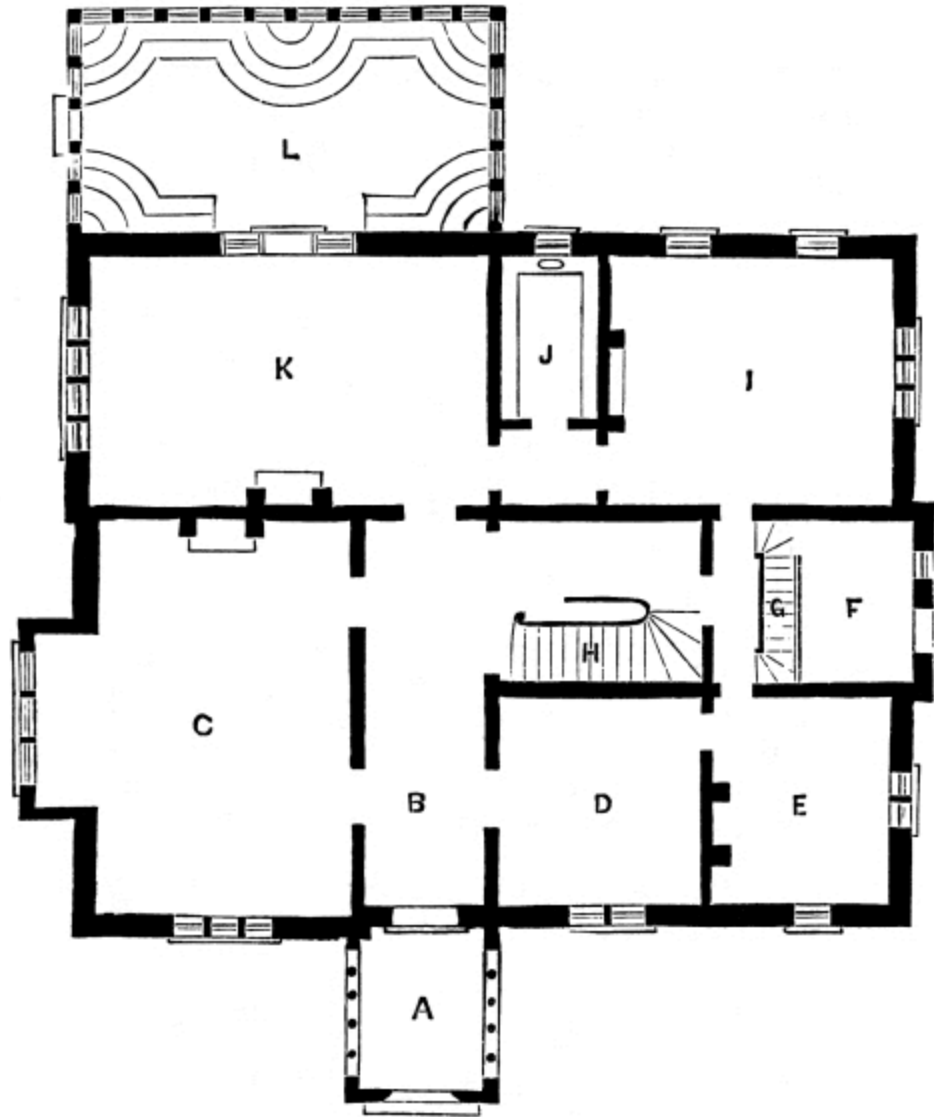
[A, The Porch](#), pierced on each side with open lights. B, the Hall, in the form of an L, and receiving light from the roof. C, the Drawing Room, with its capacious bay window. D, a Parlor. E, Library and Study. F, Side Hall, with door, under stairs, communicating with passage leading to study; (or, there may be a door opening directly into the study from the side hall.) G, Private Stairs. H, Principal Stairs, under which is a door communicating with the passage to study. I, the Kitchen. J, Pantry. K, the Dining Room, with glass door leading out into the Conservatory L.



SUBURBAN RESIDENCE.

Few arrangements of plan can be more complete. Chimnies all in the inner walls retain the whole of the heating within the house in winter. And so thorough is the natural ventilation, by doors and windows, that coolness is secured in the summer time.

Executed in stone, either hammered or rough rubble, with cut-stone trimmings, this house would present a pleasing appearance. In pressed brick, with stone trimmings, though not so consonant to surrounds of shrubbery as in stone, it would yet be a neat object and tend much to the embellishment of the outskirts of a city or village.



DESIGNS FOR SMALL CHURCHES.

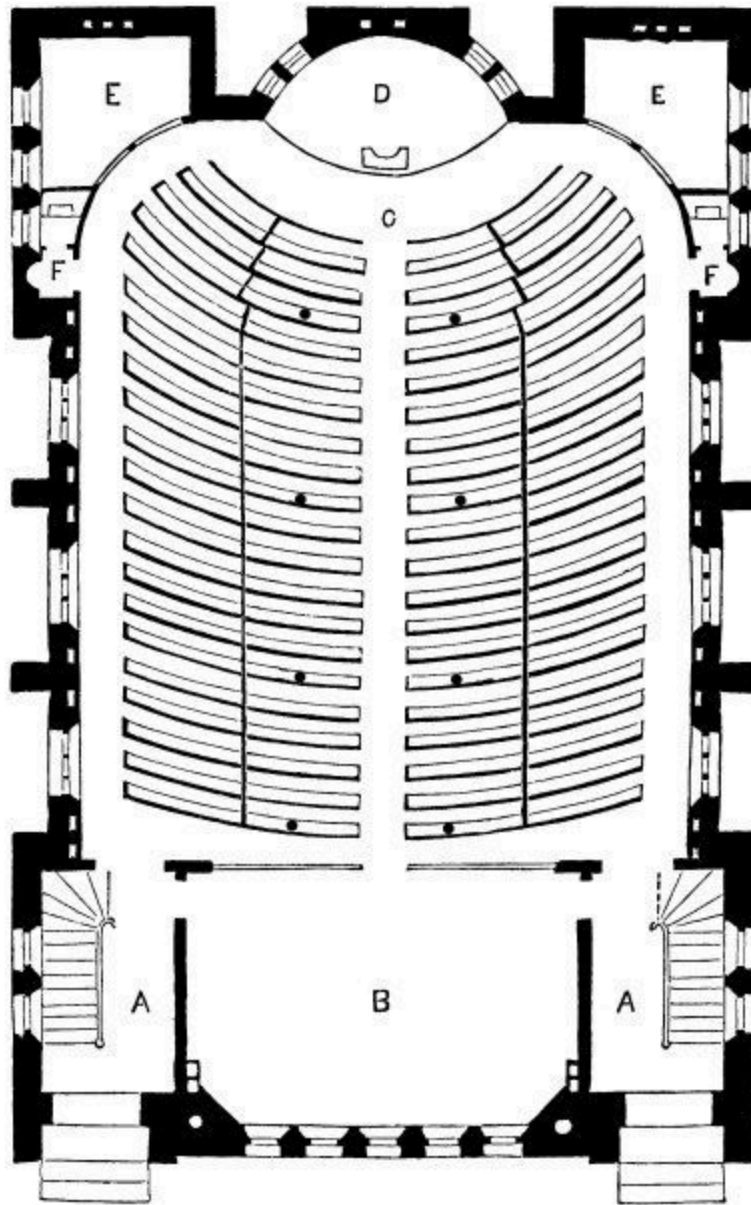
There is a great want of suitable designs calculated to meet the tastes and necessities of those communities whose funds are too limited to admit of anything approaching to architectural display. Our object, therefore, in presenting the two which illustrate our remarks, is to show the way to others to do likewise.

Churches of large dimension and assuming appearance call forth professional skill, because the expenditure will be commensurate with the

expansive ideas of the wealthy for whose benefit such edifices are constructed. But a plainer class of erections, as much wanted, should draw out the efforts of our brethren, if only for the good they may do.

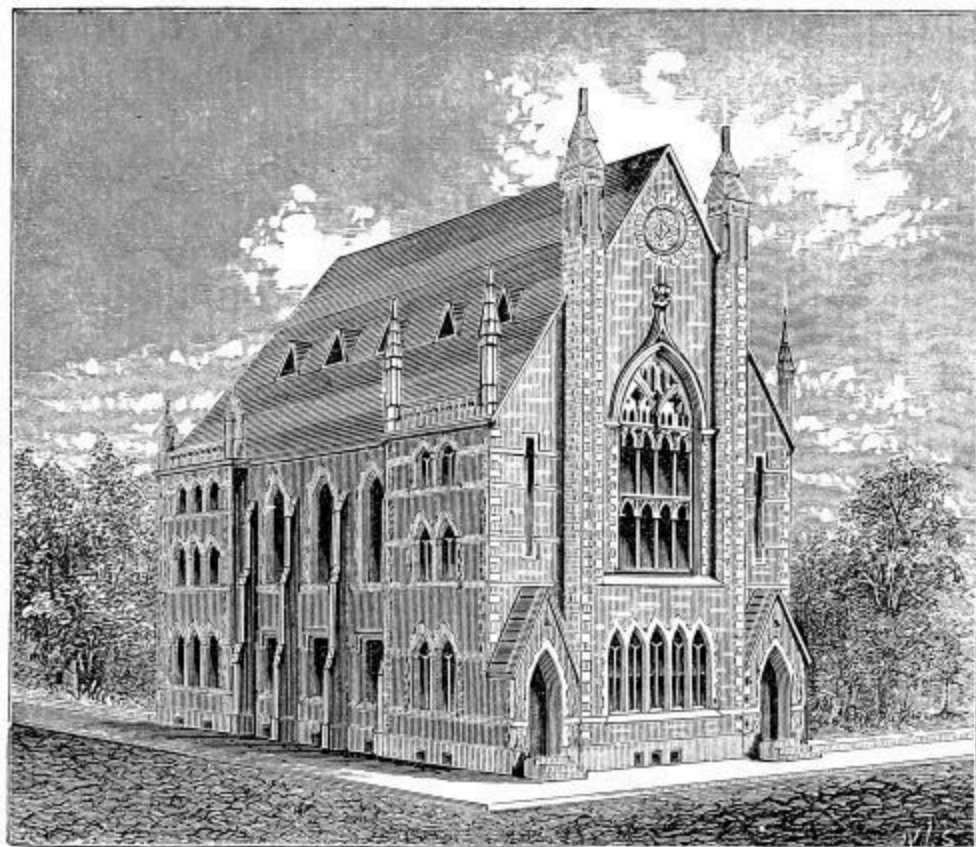
There are few architects who are not subject to the often occurring claims on their donative services in behalf of poor congregations, and, we say it with pride, that we have yet to hear of the first instance of those claims not being promptly attended to by even the busiest of our brethren. Although it too frequently happens that their liberality is severely and most thoughtlessly taxed; for there generally is in every community some spirit too restless to cease troubling even those whose time is very limited. In a serial like the ARCHITECTURAL REVIEW there is an opportunity presented to give, from time to time, sketches and instructions, by which the wants of the bodies we allude to may be met. The pastor in the backwoods, and the minister on the prairie, as well as the servant of God who teaches the poor in our crowded cities, and skill are freely given, not to them personally, but to the sacred cause they are supposed to have an interest in. But let that pass.

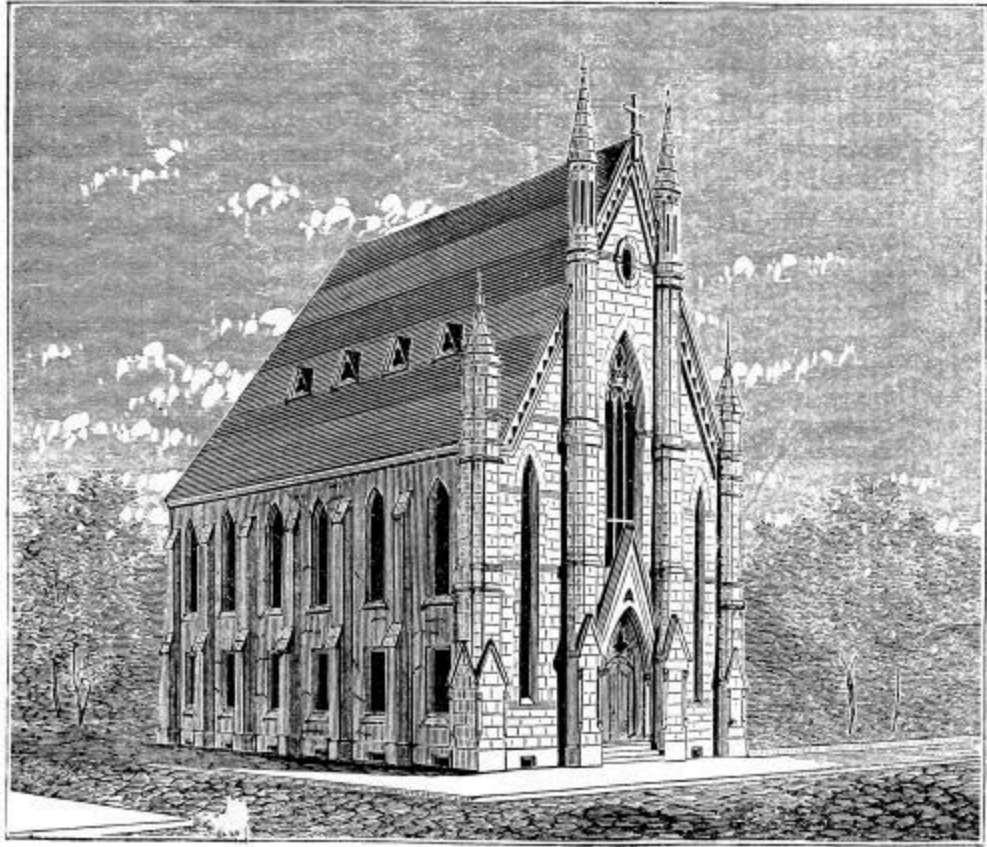
FIG. 1.



The illustrated works on Ecclesiastical Architecture, which come from the press, usually treat of a class of edifices altogether beyond the reach of the congregations whose means are limited—will each and all be benefitted by the information given, and a truly good work will thus be done. The two small churches here presented are now in course of construction in this city.

The one on the upper part of the page is a Chapel of Ease to the [Calvary Presbyterian Church](#), now building on Locust street, west of Fifteenth street.



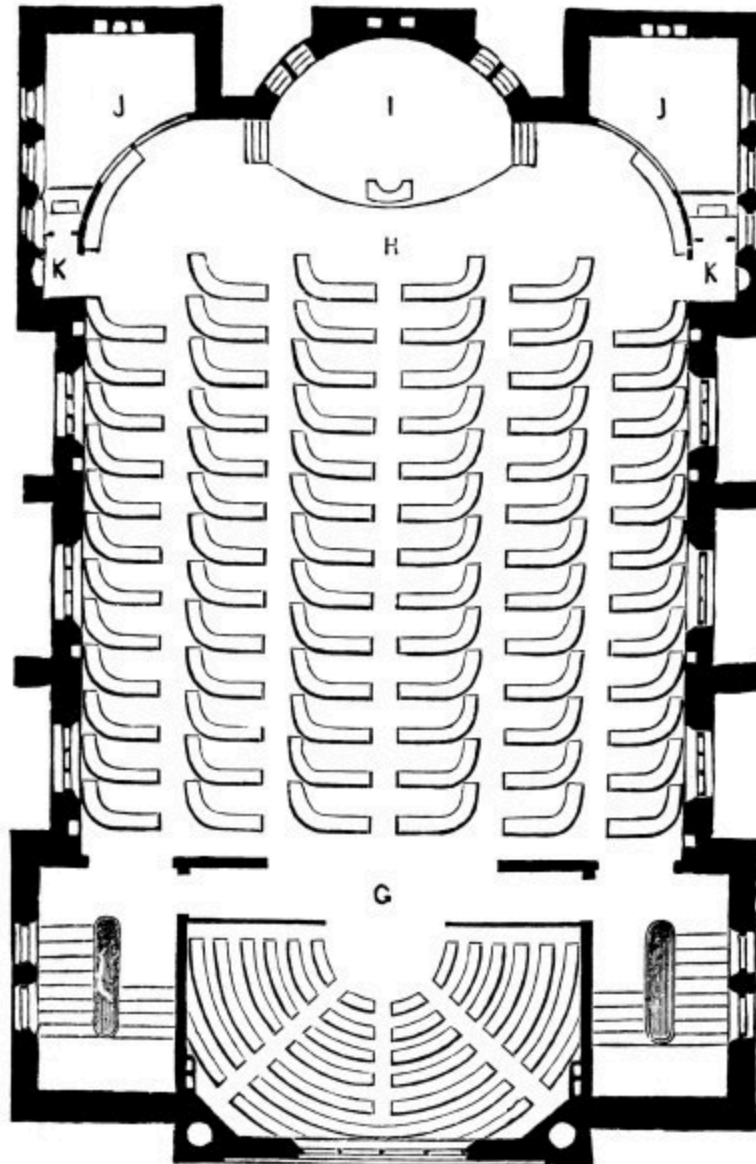


TWO DESIGNS FOR SMALL GOTHIC CHURCHES.

Its dimensions are fifty-seven feet front by ninety feet deep, outside measurement. It will be two stories high, with gallery.

The first story will be sixteen feet from floor to floor. This is to be the Lecture Room. The second story will be twenty-five feet at the walls, and thirty-nine feet to the apex of the ceiling in the centre. The Gallery will be six feet wide along the sides, circular on front, and the ends curved at the rear. Its floor will be level.

FIG. 2.



Besides the Lecture Room, the first floor will contain two class rooms and the ladies' parlor. Immediately over the Lecture Room, and of the same size, will be the Sunday-school Rooms. And over the ladies' parlor there will be the Infant School.

On the gallery are three class rooms on the front, two of which are over the Infant School Room, and one over the eastern stairway. There are two class rooms in the rear. The walls will be of rubble masonry. As high as the level of the first floor, and projecting two inches, with a wash, the exterior will be hammer-dressed. Above that, the superstructure will be all laid

broken range, pointed off, except the rear wall, which will be rubble with rock face. The whole will be faced with Trenton Brown Stone.

All the dressings of the doors, windows, buttress, caps, cornices, pinnacle caps, etc., will be distinguished by a finer class of work.

The roof and its dormers will be covered with best Blue Mountain slate, of medium size, varied with green and red color.

The interior as well as exterior finish will be Gothic in style, inexpensive yet expressive.

[FIG. 1.](#) The plan of the Lecture Room is here shown: A, A, the entrances, with stairs in each, leading to School Rooms and continuing to Gallery. B, Ladies' Parlor. C, the Lecture Room. D, Platform and desk. E, E, Class-Rooms. F, F, Water-Closets.

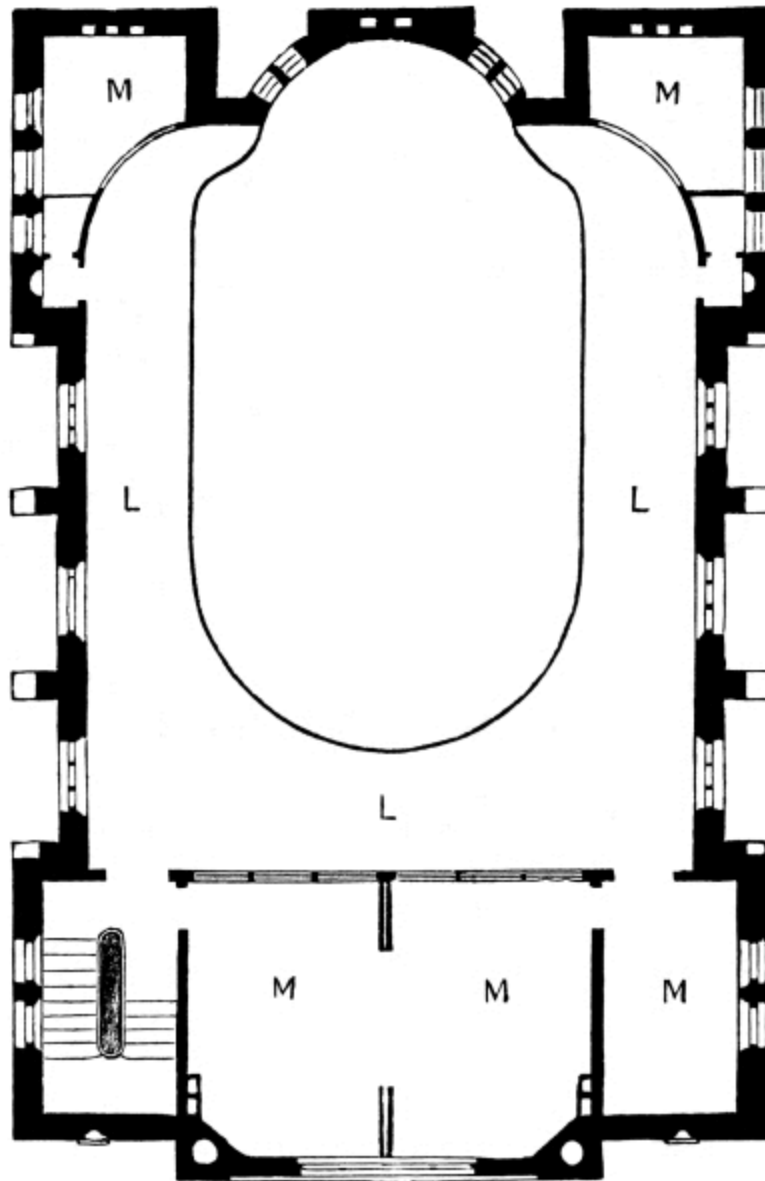
[FIG. 2.](#) This is the arrangement of the Second story, which contains: G, the Infant School Room. H, the School Room. J, J, Class Rooms. K, K, Water Closets.

[Fig. 3.](#) L, L, L, the Gallery. M, M, M, Class Rooms in front. M, M, Class Rooms in rear. It will be seen that, by means of sliding glass partitions, each floor can be considerably enlarged in accommodation. There are nine class-rooms, and school room for over six hundred children. The galleries will hold two hundred and fifty.

The illustration below that of Calvary, is the design of the [TRINITY REFORMED CHURCH](#), now being erected on the east side of Seventh street, south of Oxford street, in this city.

It is also Gothic in style, and although smaller than that just described, will, nevertheless, be a very convenient and tasteful church, and well suited to the wants of its growing congregation.

FIG. 3.

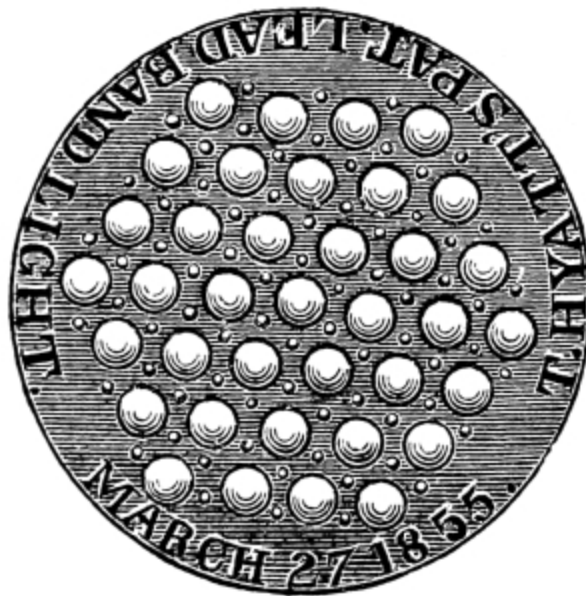


HYATT'S VAULT LIGHTS.

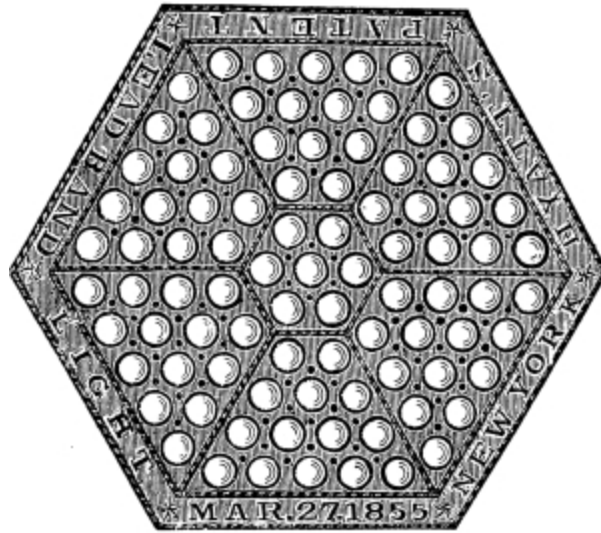
Few patents have conferred a greater blessing on society than that of which the [accompanying cut](#) is an illustration. The misery which was closely akin to area gratings, as used in “our grandfather’s day,” may yet be remembered by some not very old readers. Then light had to be admitted from the sidewalk without trespassing on the right of way by encroachment, and the manner in which that object was attained was by the use, invariably,

of open iron gratings, which, whilst they admitted the light in *bar sinister*, as our heraldic authorities would say, did not offer any opposition to the falling dirt of the street which resolved itself alternately into dust or mud, according to the relative condition of the weather. The very palpable consequence of such a state of things was, that all areas under sidewalks were an accumulative nuisance which had to be borne if day-light was desirable in underground places.

Let us pause for a moment to mentally look back on those days of dirt-clad cellar windows, if it were only to enhance the value to our mind of the present state of things.

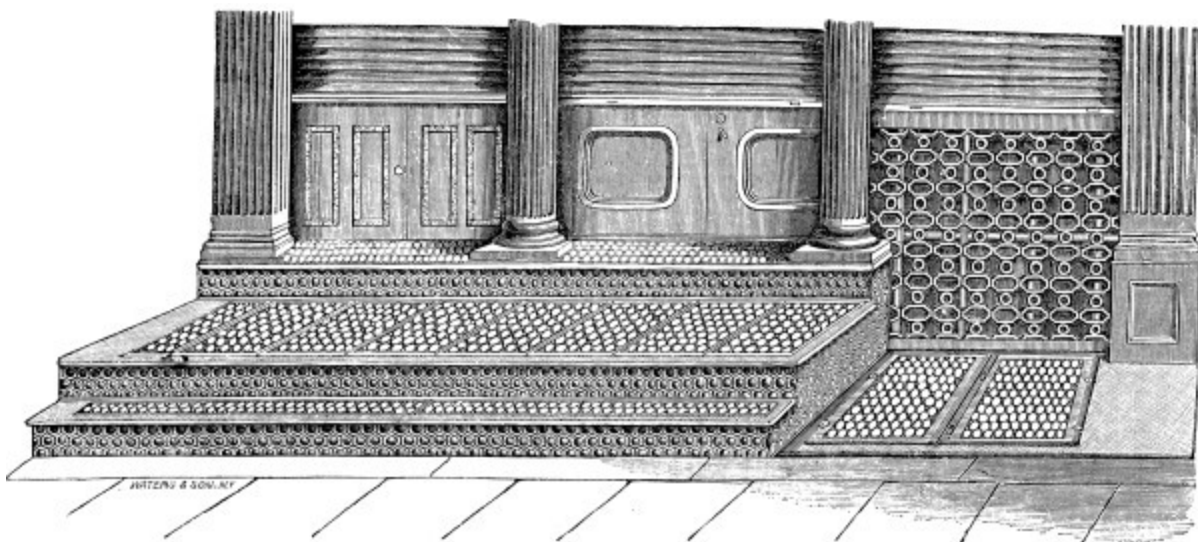


Hyatt's Patent Vault, and Side-walk lights, are so well known and so universally appreciated North, South, East, and West, now-a-days, that it is doubtful whether we are enlightening a single reader of the REVIEW in thus alluding to them. But, unfortunately there are people so listlessly unobservant in this world of ours, as to walk over them, aye, and walk under them, without perceiving the benefit enjoyed from them. Such people look on all improvements without wonder or admiration, and calmly set them down as matters of course—things that were to be, improvements—the growth of necessity. The inventive mind that gave them birth is neither thanked nor thought of. But all men are not so stolid. Many will take an interest in the benefaction and the benefactor, and to such the present notice will recall a duty—the grateful acknowledgment of a benefit bestowed.



The sidewalk lights are powerfully strong as well as perfectly weather-proof and they can be turned out in any required form in single plates to a maximum size of six and a half feet long by two and a half feet wide, or in continuous platforms. They are likewise made to answer an excellent purpose as steps and risers, or even as entire flights of stairs of any desired length. They are three quarter inch thick, hexagonal shaped glass, well secured and presenting a really handsome appearance.

In our preceding number we made some observations on a more fitting system of awnings than that now in use.



We think there can be very little doubt but this very invention could be well made available for such a purpose, and we sincerely hope that the hint will not be lost sight of.

BROWN BROTHERS of Chicago have for the last ten years been active in the manufacture and sale of the patent sidewalk lights, and there is scarcely a city of any pretensions in the Great West that has not awaked up to the use and value of this most beneficial invention, and the pleasing consequence is that the Messrs. B. are now doing an immense business in the manufacture of them, at 226 and 228 Monroe street, Chicago, where the orders of our friends the Architects and Builders who propagate improvements in the growing cities of the irrepressible West, will be attended to, with that promptitude which has hitherto made the name of the firm of BROWN BROTHERS so well known, and their excellent manufacture so fully appreciated.

WHITE LEAD BY A NEW PROCESS.

The manufacture of this important and useful pigment has been very successfully prosecuted within the past year, by a new process, the invention of Dr. H. Hannen of this city, and is destined to supersede the old method, both as regards economy in preparation and purity of material. The old or Dutch process, requiring some six to eight months for its completion, fit for painter's use; while by the Hannen patent it can be produced in from ten to fifteen days. The quality of the article is said to be fully equal, if not superior, to that of the lead made by the old method. The process of manufacture, as far as we can learn, is as follows:

The best Spanish pig lead is melted in a large iron kettle, holding from fifteen to eighteen hundred weight, and then drawn off by a suitable valve, and allowed to run over a cast-iron wheel or drum, about six inches on the face and three feet in diameter, running at a high speed, and kept cool by a stream of cold water constantly playing on it. The lead, in passing over this wheel, is cast into ribbons about the thickness of paper, it is then taken and placed on lattice shelving in rooms some eight to ten feet square, made almost airtight by a double thickness of boards, and capable of holding some three tons of the metallic lead as it comes from the casting machine in ribbon form, the temperature of the room is then raised by injecting steam

to about one hundred degrees, and then sprinkled several times a day with diluted acetic acid, converting it into sub-acetate or sugar of lead. While this operation is going on, carbonic acid gas is forced into the room by means of a blower or pump, which decomposes the acetate and forms a carbonate of lead; this operation of forming an acetate, and then a carbonate, requires from five to six days, until a complete corrosion of the lead is effected; the room is now allowed to cool and the lead to dry, after which it is taken out and sifted through fine wire sieves, which separates all undecomposed lead or other impurities. It is then ready for washing and drying. The finely powdered lead is mixed with water into a thick pasty form and ground in a mill of similar construction to an ordinary flour mill, from which it is allowed to run into large tubs filled with water, and thoroughly washed and allowed to settle. The last or finishing operation is to place it in large copper pans, heated by steam, when it is dried; from thence taken to the color grinder, where it is mixed in oil ready for the painter's use.

PAINTERS AND ARCHITECTS.

There is a presumptuous feeling in the breasts of those who, *par excellence*, assume the style and title of “Artists,” both in the Old and the New World, which it would be well to look into were it not that valuable time might thus be wasted on an exceedingly contemptible subject. We allude to the arrogation of eminence by those autocrats of the easel, who, not content with the undue position conceded to them by the vain and the frivolous who stilt themselves on their recognition of “high art,” and affect to govern the very laws of taste itself, go farther in the fulness of their ambition, and seek to ignore ARCHITECTURE as an art. This outrage on common sense is not confined to America, it has been continuously practised, if not boldly promulgated, for over a century in London, by an institution bearing the absurd title of THE ROYAL ACADEMY, originally intended to foster and advance the interests of Architecture, Painting, and Sculpture, yet in forty elections, or rather selections, of Associates, that is, of those ordained to emblazon their names with the R. A., *but four were Architects!*

And, notwithstanding the studious efforts made by our profession to elevate our position and draw at least our share of public attention, we find that this Royal Academy and the rest of the aristocratic Dundrearifications, positively prohibit the appearance of architectural designs upon the walls of their National Galleries by crowding every available foot of wall space with easel-work, (we beg pardon—“paintings,”) ephemeral, unnatural, mannerized exudations of the “modern school,” that barely patronizes Nature as a stupid fact, which to be got round must be obliterated in gaudy coloring. But, shall Architects make bold to criticize these “Artists?” No, PAINTING is a sublime gift, by the magic touch of which the coarse inelegant canvas is made to put forth emanations of the etherial mind, which it were a pity to limit to the paltry boundary of a gilded frame!

What is ARCHITECTURE?

Where would the art of PAINTING find a shelter, were it not for Architecture?

Do the gentlemen of the brush and palette ever look around and above at the walls, the ceilings, or even at the tessellated floor of the rooms where their small framed efforts are on exhibition, and suffer their overweening vanity to acknowledge that ARCHITECTURE is really something?

How many painters can properly depict it? How many?

The ignorance which urges the pre-eminence of PAINTING at the expense of ARCHITECTURE is more to be pitied than condemned. And the public patronage lavished on the one and withheld from the other, is superinduced by the ease with which any one can assume to be a critical admirer of an art whose governing rules are imaginary rather than real or substantial.

Some see beauty in the fidelity which a painting bears to Nature. Others consider that very fidelity as slavish imitation. And a very general notion obtains amongst painters of “assisting Nature.” Now, ARCHITECTURE stands upon the solid base of TRUTH. Without imitating, it borrows applicable ideas from Nature to be used in carrying out its designs. Nor is it merely the imaginations, limnings, as in the case of PAINTINGS; those designs have to be executed. CONSTRUCTION then comes in as the solid, tangible, work of art, which shall defy the elements and render ARCHITECTURE the protectress of PAINTING, without whose solid enduring defence the more fragile art would speedily decay and become unknown.

But, are not the professors and admirers of ARCHITECTURE themselves to blame for the degraded position it holds to-day as an art, here and in Europe? Why is there not more practical enthusiasm, and altogether less contemptible jealousy, and ill-natured feeling, amongst all who claim to have an interest in this the grandest and most over-shadowing of the Arts?

If PAINTING must needs hold an exclusive position as regards the public exhibitions of what is most erroneously called the “Fine Arts,” why cannot ARCHITECTURE and SCULPTURE assert their dignity, and give the public a chance to patronize them independently? The truth is that Architecture and Painting do not at all agree in sentiment; the one is a mere luxury, and no more; the other is a necessary art, adorned or unadorned. The one can be glanced at and instantly understood; the other demands the effort of the

mind to study and to comprehend. In PAINTING, the eye is the arbiter; in ARCHITECTURE, the eye and the mind must form the judgment. It is not what a merely pretty picture is displayed; it is—how would that design look in execution?

Most of people who go to a “Fine Art Exhibition” are superficial observers. They glance at pictures by the hundred. Such are not the persons from whose judgment ARCHITECTURE can expect even a recognition. They have been bedazzled with the sheen of the gilded frames, and the well laid-on varnish which bedizens the bright pigments of the gaudy glare of Art, which they have just left, and are, of course, impatient of the more staid and methodical elevations or perspectives, now presented in a narrow crowded section to their view. They have not time nor inclination to pause and consider them. They cannot bear to lose the impressions made by the “sweet shaded alley,” the “dancing streamlet,” or the “green reflective lake,” with that charming sky that looks so much more like heaven than nature. No, it will not do to exhibit ARCHITECTURE and PAINTING together, and it is time to acknowledge this so often proven fact. The two must be distinct. Let Architects put forth their powers, and show the community what their Art really is, and what it is capable of. People will go expressly to view an exhibition of Architectural designs, combined with Sculpture, and take much pleasure in the visit, because their mind is prepared for the occasion, and will not be distracted by a rival exhibition of quite another effect. To say that the public generally will find no pleasure in the consideration of Architecture is to assert that which is disproved by fact. When the Commissioners, appointed to choose a fitting design for the new Post Office at New York, threw open to a limited number of visitors the inspection of the collection of designs, the rooms were crowded each day of the exhibition, and innumerable applications were made for tickets of admission. Had all the public been allowed the privilege, no doubt it would have been universally accepted. Yet that was but a very uninteresting display compared to one in which the subjects would be manifold, and the scales various. Not to speak of the freedom of display in color, which on the occasion adverted to was necessarily confined to an extreme limit.

Why cannot our Architects have an independent exhibition? There is nothing to be gained, but on the contrary every thing to be lost by clinging to the skirts of the *painters*. An effort in this direction could not fail to meet

with the warmest support from our monied citizens, who are constantly proving substantially their regard for the progressive welfare of Architecture, by expending vast sums in buildings. And we have no doubt, but that State Legislatures would promptly and liberally aid any such effort to educate the general public in an art so intimately connected with the history of civilization.

HONOR TO WASHINGTON.

The anniversary of this great nation's independence never was more fittingly honored than on the Fourth of July last, when, in this city, and in the front of the glorious old Independence Hall, Philadelphia inaugurated her statue of him who was FIRST IN PEACE, FIRST IN WAR AND FIRST IN THE HEARTS OF HIS COUNTRYMEN. There is not in the United States a single spot more sacred to the cause of Freedom than that on which stands Independence Hall, where our great fathers of the Revolution so nobly pledged to the cause of mankind their lives, their fortunes, and their sacred honor, and where the truly noble Washington was heard and seen, when the hopes of an embryo nation rested on his integrity.

Although the thought well suggests itself that an honor such as that just now paid the great patriot's memory should long ere this have been credited to Philadelphia, yet it is never too late to do our name justice before the world; and it is appropriate that the rising generation of a closing century should thus mark the establishment of a free government for which he fought and conquered.

Thanks to the school children whose contributions thus have given to Philadelphia, what their sires so long neglected, a testimonial worthy of our grateful recollection of the foremost of Americans.

On the 13th of December, 1867, a contract was made with our eminent citizen artist, Mr. J. A. BAILEY, and on the 2d of July, 1869, the material for the granite base was delivered on the ground. The following day the statue was duly erected, where it now stands in front of the entrance of that venerated Hall.

In the centre of the foundation is placed a box containing the names of children and teachers, Directors and Board of Controllers, Mayor and City

Councils, heads of departments, records of the Association, etc., and a copy of the Holy Bible. The base of the statue is of Virginia granite, from the Richmond quarries, and is in four pieces, weighing about twenty tons. The statue is of white marble, 8 feet 6 inches high. The left hand of Washington rests on the hilt of his sword, sheathed in peace; his right hand rests on the Bible, the Bible on the Constitution and American flag which drapes the supporting column on the right of the figure. The weight of the figure is about six tons. The whole height of base and statue is 18 feet 6 inches. On the north front the base will bear the name—WASHINGTON; on the south, this inscription:

ERECTED
BY THE
WASHINGTON MONUMENT ASSOCIATION
OF THE
FIRST SCHOOL DISTRICT OF PENNSYLVANIA.

The total cost, including a railing, will be about \$6,500.

The ceremony of the unveiling was a most impressive one, the children being in the act of singing “Hail, Columbia,” when, at a given, signal, the flag covering the noble statue was raised, and from its folds came forth innumerable small flags which flew among the people and were eagerly caught.

As the marble image of Washington came into view the cheers of the assembled thousands were only outvied by the cannon in the square, and the national hymn was for the time drowned in the enthusiasm of the event.

The President of the Washington Monument Association Mr. GEORGE F. GORDON, in an appropriate address to the Mayor and Select and Common Councils, presented the beautiful monument to the city. It was received by the Mayor, Hon. DANIEL M. FOX, in a suitable reply, and the benediction being pronounced, this most interesting event became part of the brightest of Philadelphia’s chronicles.

The munificence of our fellow-townsmen, W. W. Corcoran, Esq., has been handsomely acknowledged by the National Academy of Design, at New York, which has transmitted to him congratulatory resolutions with reference to his recent foundation of a gallery of art in this city.—*Washington Chronicle*.

NEW SOUTH WALES.

Our latest files are to April 21st, inclusive. Sydney was at that time in high spirits over the recent visit of the Prince Captain of H. M. S. *Gallatea*. The most noteworthy action of whom was the laying of the corner-stone of the testimonial to the hardy navigator and discoverer, Captain Cook. We extract the remarks of the leading journal of Sydney.

“THE CAPTAIN COOK MEMORIAL.—A monument to the memory of Captain Cook will be rather an expression of our admiration for his character and services than an enhancement of his fame. The last generation was filled with wonder at the narrative of his discoveries. The first quartos that record them display in most striking forms the scenes and objects he made known to the world. He visited many islands of the Southern seas, whose voluptuous and animated social life attracted as to a new-found Paradise. Subsequent experience scattered the illusions of fancy, but brought out more clearly the value of his labors. New South Wales presented to his view a land of savages, lowest in the scale of civilization, but it also offered a noble field for British colonization, perhaps less appreciated while America was still a dependency of England, but brought into notice a few years after that country ceased to belong to the Crown.

“COOK first landed at Botany Bay, on the 19th of April, and on the 23d of August, he took possession of the entire country in the name of the SOVEREIGN of England. The precise spot where he anchored is marked in the charts by a nautical symbol, and can thus be identified. On reaching the shore he found a spring of water ample for the wants of the ship, and tradition has reported that he bent his knees in adoration of the Supreme Being.

“The character of COOK as a navigator occupies the first rank in nautical sciences. It is to his high honor, that modern times, though they have added to his discoveries, have been rarely able to dispute them. Nothing is

superfluous—nothing is obscure. The modern investigator starts from the observations made by COOK as undoubted facts. Every year displays more strikingly, not only the results of his discoveries and their value, but the almost prophetic foresight which presided over them.

“The history of Captain COOK is an example of the lofty position which may be taken by the humblest ranks when attended with high intelligence and superior moral qualities. The first step of his naval career was as a cabin boy. He rose to the command of an expedition which was suggested by scientific men, and their warmest hopes were more than fulfilled. They had seen with regret the blanks in the map of the world, and the ignorance which prevailed in reference to the true character and capabilities of countries partially known. The men of science who accompanied him on his voyage acquired for a time a scarcely inferior fame. Mr. BANKS and Dr. SOLANDER are names familiar to the readers of COOK’s Voyages, but the magnificence of his achievements leaves in the shade every inferior merit. He stands forth as the founder of a new era in nautical discovery, and as the revealer of a new world.

“Could Captain COOK have seen the spot on which it is proposed to erect his monument, and from thence, with superhuman knowledge, anticipated the events of this day, he would have been overwhelmed with awe.

“EDMUND BURKE delineated, while the struggle with America was still transpiring, the emotions of astonishment with which he supposes Lord BATHURST, then an aged statesman, might in the days of his youth have looked forward, under the guidance of some celestial instructor, to the events which had raised American colonization from insignificance to greatness. But what emotions would have stirred the heart of COOK, if, standing on this spot, he had foreseen the progress of colonization, the painful labors included in the first fifty years, and the immense prosperity of the last.

“Had such heavenly anointing enabled him to foresee all this, his grateful spirit would have been filled—with—what sacred joy! Still further extending his intellectual prospect, he might have foreseen the arrival of a vessel furnished with the results of science then unattained, advancing like some being, instinct with intelligence, from port to port, through billows

over which he was tossed, and independent of winds for which he had to wait, arrived at a fixed hour at the haven of its destination. And still farther, he might have seen the great grandson of that monarch whose name he proclaimed as the lord of this territory—the son of a royal woman who has inherited all the virtues of her race, without its faults; and he might have seen that son, surrounded with a multitude of her subjects, standing over the first stone of an edifice to do honor to his memory.”—*Sydney Morning Herald*, March 27.

“THE NEW POST OFFICE, SYDNEY.—The keystone of the central arch of the new Post Office, George street, was laid by His Royal Highness the Duke of Edinburgh, on the 1st instant, in the presence of a vast concourse of spectators. A large platform was erected behind the arch, and on a level with the stone, access to which was obtained by carpeted stairs, springing from the northern side of the building.

“The stone laid by the Prince forms the keystone, archivolts and two spandrels of the central archway of the George street front. Upon the face are to be carved the Royal arms, and upon the coffered soffits the arms of the Duke. The dimensions of the stone are:—Length 13 feet 6 inches, width 4 feet 6 inches, and height 6 feet 6 inches—the whole being equal to 394 cubic feet. The weight is twenty-six tons. This stone is doubtless the largest yet laid by his Royal Highness, and it is probably the largest block of sandstone he will ever lay, for it would be difficult, if not indeed impossible, to get *sound* blocks of sandstone of equal size from any quarry in England, or elsewhere. Few cities are so favorably situated for sandstone as Sydney, for in almost every direction blocks of this description of freestone may be obtained of almost unlimited dimensions, and without a flaw. The most casual observer of the new Post Office cannot fail to notice the massiveness of the stones used in the building, and the solidity of the structure is unequalled by any other erection in the city. The contractor has placed very powerful cranes in his quarries at Pymont, whence these immense blocks of stone are obtained, and great credit is due to Mr. C. Saunders for the workmanlike manner in which these blocks—far exceeding in size anything previously attempted in the colony—have been quarried. The difficulty of removing these heavy blocks of stone must be very considerable; and the stone laid by the Duke of Edinburgh was equal to the force of twenty-one horses, calculating a horse to draw about twenty-

five cwt. Ordinary wagons or trucks usually carry weight not exceeding 5, or, at most, 6 tons; and as there are in this building many blocks of granite and freestone of 10 to 20 tons, the difficulty of carriage can easily be seen. In hoisting and fixing these large stones ‘travellers’ are used, which can move longitudinally and crossways; and as the lift is directly over the stone to be fixed, there is less liability of accident than by the use of cranes or other contrivances.

“The building progresses as rapidly as the elaborate nature of such work will admit. It is now to the height of the first story, twenty-five feet from the floor line, which is three feet above the causeway in George street. The works are being carried on under the superintendence of the Colonial Architect, Mr. James Barnet. The contractor has fixed all the polished granite columns on the work front facing the street, which is to be taken through from George street to Pitt street. They are exceedingly beautiful, and are resplendent with a lustre brighter than that of marble. The polish has been brought out by an elaborate process, and is, we believe, ineffaceable by atmospheric influences. Each column is polished by machinery—inconstant friction continued for a fortnight being requisite to bring out the lustre. There are to be twenty-seven columns in the George street front, which the Government have also decided shall be of polished granite, material which for beauty and durability cannot be surpassed even in Europe. The building, when completed, will compare favorably with any structure erected for a similar purpose elsewhere.

“The blue granite used in the edifice is obtained by Mr. Young from his quarries at Moruya, about one hundred and sixty miles to the south of Sydney. The quarries are opened in the side of the hill—a mountain of granite in fact—and about half a mile of railway constructed across the swamp carries it to a granite jetty, which has been built in the river, into water deep enough to admit of vessels drawing fifteen feet of water loading alongside. The granite is sound—sufficiently so, indeed, to admit of two hundred feet lengths being quarried. A block has been got out for the front columns of the Post Office, which weighs nearly three hundred tons, and the dimensions of which are:—Length, 22 feet; breadth, 22 feet; thickness, 8 feet; total contents being 3,520 feet.”

BUILDING IN CONCRETE.

It is something to be wondered at, the slowness with which the advantages of concrete, as a building material, have been developed and accepted by practical men. As a foundation it is beyond all doubt the firmest, simplest, and most economical. But, its merit is not confined to underground operations; for, as has been repeatedly maintained during the last twenty years, it is capable of making walls of unsurpassing strength and durability, giving comforts which no other material will. It is true that certain parties have sought to astonish the world with securely patented *inventions*, by which Nature's humble efforts at making granite were at once surpassed, and the old foggy way of the consolidation, by the tedious action of time, of grains of mica, quartz, and feldspar, set aside by the use of this invaluable mode of making as good an article with one man power at a rate fully equal to supplying the demands of all who want stone houses erected rapidly from the raw material!

All this is arrant folly, and should not be listened to, much less patronized. The making or undertaking to make stone in blocks is a step, aye, a long stride backwards.

The object of cementing together blocks, whether of brick or stone, is simply to produce one solid mass. And it is because we cannot conveniently carve out in a *monolith* or mass together in one *tumulus* the desired dwelling or temple, that we are forced either to break blocks of stone into fragments, or mould and burn earth into bricks. Now the idea of forming artificial stone into blocks still leaves the expensive necessity for cementing them together; and therefore instead of improving our condition, actually leaves us worse off, by giving us, as a substitute for Nature's well-tested material, a most unreliable article, which has already too clearly proved its utter worthlessness. However, this should not cause the friends of progress to give up all idea of simplifying and economizing the mode of wall structure. On the contrary it should stimulate them to make that exertion in the right way, which has hitherto been so persistently and blindly made in the wrong.

In Europe they are taking this subject into serious consideration. In England, under the name of CONCRETE; in France, under the title of BÉTON. In the latter country, much has been done lately, and all arising out of the excellent work on cements given to the world by M. VICAT, whose name should be enshrined forever in the Temple of Fame, for the amount of good,

present and prospective, which his earnest labors have done the Art of Building.

One of the most indefatigable and successful of experimenters in *béton* is M. COIQUÉT, who has proved beyond all cavil the excellence of that composition when applied to the sustaining of weight or resistance of pressure.

In London we find Messrs. Drake, Brothers and Reed, under Her Majesty's Letters-Patent, undertakers of Building in Concrete.

It is the machinery they use that is patented, we believe, and not the material; for there are many others in this branch of business. Mr. JOSEPH TALL, of London, has also a patent for a peculiar method of building in concrete, and has executed some contracts in Paris, where, in 1867, he took a prize at the Exposition.

It is evident, then, that concrete is forcing its way, and that it is not an unworthy subject for the inventive minds of our astute countrymen.

What we particularly need in order to give an impetus to construction in concrete is a well-systematized apparatus, movable and always available, and that men should be drilled to work to the greatest possible advantage; for it is the want of these requisites that makes concrete to-day a material so little known and so seldom used.

Let an active company, with sufficient capital, start the business in any of our large cities, and concrete will soon assert its excellence as a building material, and an investment will be secured, giving profit to its holders and satisfaction to a very large section of our population, to whom economy must prove the key to comfortable independence.

The quarry companies in Connecticut were never doing a heavier business than this season. Three quarries now employ over one thousand laborers, seventy-five horses and one hundred yoke of cattle.

A REMARKABLE CENTENARY.

How few there are who pause for one instant from their plodding after the deified “Dollar,” to reflect that this present year, 1869, is the most remarkably commemorative of any yet on the Book of Time.

It is now one hundred years since HUMBOLDT, CUVIER, the first BRUNELL, JAMES WATT, Jr., and Sir THOMAS LAWRENCE, among the most eminent of the world’s civilians—and NAPOLEON the First, WELLINGTON, SOULT and NEY, among the most advanced rank of mighty military chiefs, had birth.

It is one hundred years since the elder WATT’s condensing steam engine was invented, and that invention which brought poverty with its production has, in these hundred years, revolutionized the globe, and made not alone individuals, but whole nations wealthy and powerful.

No nation on the globe owes more to WATT’s steam engine than does this of ours. Where now would Civilization be coiled up? Where now would Science be secluded comparatively unnoticed and unknown—were it not for that one invention?

The peoples of the world have been growing and multiplying, and where would have been the room, or the employment for the teeming millions, were it not for that happy thought which in 1769 became a palpable fact?

A wise Providence was over all, and the brain that worked out the idea of the condensing steam engine was but doing its special part in the great work of civilization and progress.

This Centenary is one which should not be allowed to pass unheeded, especially now that we have just drawn the extremes of the earth nearer, not alone to the ear, but to the eye itself.

“How fast they build houses now!” said H.; “they began that building last week, and now they are putting in the lights.” “Yes,” answered his friend, “and next week they will put in the liver.”

AUTOMATIC WATER ENGINE.

An important discovery connected with the raising of water is claimed to have been made by Dr. Bouron, a physician of some reputation, residing at Heverville, Seine Inférieure. It appears that by a very simple piece of mechanism he can raise a continuous stream of water to almost any altitude, without labor of any kind, and without expense, beyond that necessary for the first cost of the machine, and this is by no means large, considering the amount of useful work which it yields. Dr. Bouron states that the power of the machine is based upon a natural and immutable mechanical principle, and that by it there may be created a continuous current of water at the surface of the soil, wherever there exists, no matter at what depth it may be, a spring of water. The machine is intended to supersede all existing pumps, its construction not being more expensive, whilst it has the additional advantage that no expense is incurred for keeping it constantly and usefully at work, although other pumps, especially when the water is raised a great height, necessitates enormous expenses compared with the useful effect produced, and that, too, during the whole time they are at work. It must not be forgotten, however, that it is a stream and not a jet of water which the new machine produces, so that, although it would be well adapted to supply water to fire engines, for example, it could not replace them. It is claimed that the machine will yield the same quantity of water as that being produced by the spring to which it is adapted, (less, of course, the loss inseparable from the working of all mechanical apparatus), and at any height, whether it be one thousand metres, two thousand metres, or more. Dr. Bouron also observes that, however paradoxical it may appear, he has found "the greater the height to which the water has to be raised the greater is the power of the machine." But the relative proportion of the power to the speed is quite in conformity with the principles of mechanics. The greater the height to which the water has to be raised, the greater the power and the speed that can be brought to bear upon it; but the greater the horizontal section of the column of water to be lifted, the more will the speed diminish.

REMARKABLE MASONIC INCIDENT.

The first masonic funeral that ever occurred in California took place in the year 1849, and was performed over the body of a brother found drowned in the Bay of San Francisco. An account of the ceremonies states

that on the body of the deceased was found a silver mark of a Mason, upon which were engraved the initials of his name. A little further investigation revealed to the beholder the most singular exhibition of Masonic emblems that was ever drawn by the ingenuity of man upon the human skin. There is nothing in the history or traditions of Freemasonry equal to it. Beautifully dotted on his left arm in red or blue ink, which time could not efface, appeared all the emblems of the entered apprentice. There were the Holy Bible, the square and the compass, the 24-inch gauge and the common gavel. There were also the mosaic pavement representing the ground floor of King Solomon's Temple, the intended tessellated pavement which surrounds it and the blazing star in the centre. On his right arm, and artistically executed in the same indelible liquid, were the emblems pertaining to the fellow craft degree, viz.: the square, the level, and the plumb. There were also the five columns representing the five orders of architecture—the Tuscan, Doric, Ionic, Corinthian, Composite.

In removing the garments from his body, the Trowel presented itself, with all the other tools of operative Masonry. Over his heart was the Pot of Incense. On the other parts of his body were the Bee Hive, the Book of Constitutions, guarded by the Tyler's Sword, the sword pointing to a naked heart; the All-seeing eye; the Anchor and Ark, the Hour Glass, the Scythe, the forty-seventh problem of Euclid; the Sun, Moon, Stars, and Comets; the three steps emblematical of Youth, Manhood, and Age. Admirably executed was the weeping Virgin, reclining on a broken column, upon which lay the Book of Constitutions. In her left hand she held the Pot of Incense, the Masonic emblem of a pure heart, and in her uplifted hand a Sprig of Acacia, the emblem of the immortality of the soul. Immediately beneath her stood winged Time, with his scythe by his side, which cuts the brittle thread of life, and the Hour Glass at his feet which is ever reminding us that our lives are withering away. The withered and attenuated fingers of the Destroyer were placed amid the long and gracefully flowing ringlets of the disconsolate mourner. Thus were the striking emblems of mortality and immortality beautifully blended in one pictorial representation. It was a spectacle such as Masons never saw before, and, in all probability, such as the fraternity will never witness again. The brother's name was never known.

NECESSITY FOR PURE AIR.

Those of our citizens who were “to the manor born,” and never left their native land, cannot form any idea of the comfort they enjoy as compared with the misery endured from birth to death by thousands of kindred humanity in the other parts of the world. Even in highly cultivated and brilliant England and her dependencies, we find enough to shock the feelings and make us ask ourselves “can such things be?”

In a pamphlet recently given to the world, DR. MORGAN, a Master of Arts, and a prominent member of the British Medical Association, repeats in print a paper which he read before that learned body at Oxford, in August last; and but for which publication we would have been in ignorance of the actual depth of misery to which so many good and faithful subjects of that proud and wealthy monarchy are condemned uncared for and unthought of.

“The author remarks that the housing of the poor, while beset with great difficulties, is most intimately connected with the future prosperity of the great mass of the people. In all our great cities, there are unhealthy quarters, where the death rate is exceptionally high, and the reason of this, after careful inspection of many such places, Dr. Morgan believes is to be found in this statement. Bad air, or too little of it, kills the people.

“Men will grow robust and vigorous, the author remarks, on very poor food, in very dirty cabins, and in very sorry attire, provided they enjoy a pure and bracing atmosphere, and the great physical development of the nations of the Hebrides and the western highlands of Scotland is cited as an example. In striking contrast to this, we find that in the Isle of St. Kilda, a small island, numbering about eighty inhabitants, three out of every five infants born alive are carried off a few days after birth by a convulsive affection allied to tetanus, the difference being apparently due to the huts having no smoke-hole in the thatch, and being rendered impervious to air by double walls filled in with peat and sods, the object of which is to prevent the escape of smoke, and in due time the soot is collected and used as manure.”

DRINKING FOUNTAINS—This philanthropic movement which offers the grateful cup of Nature’s refreshing beverage to the parched lip of the

passenger, is one that takes a high place indeed in the church universal, at whose shrine all bend in unison, and know no discordant thought, but love one another for the love of God.

LESSONS FOR LEARNERS.

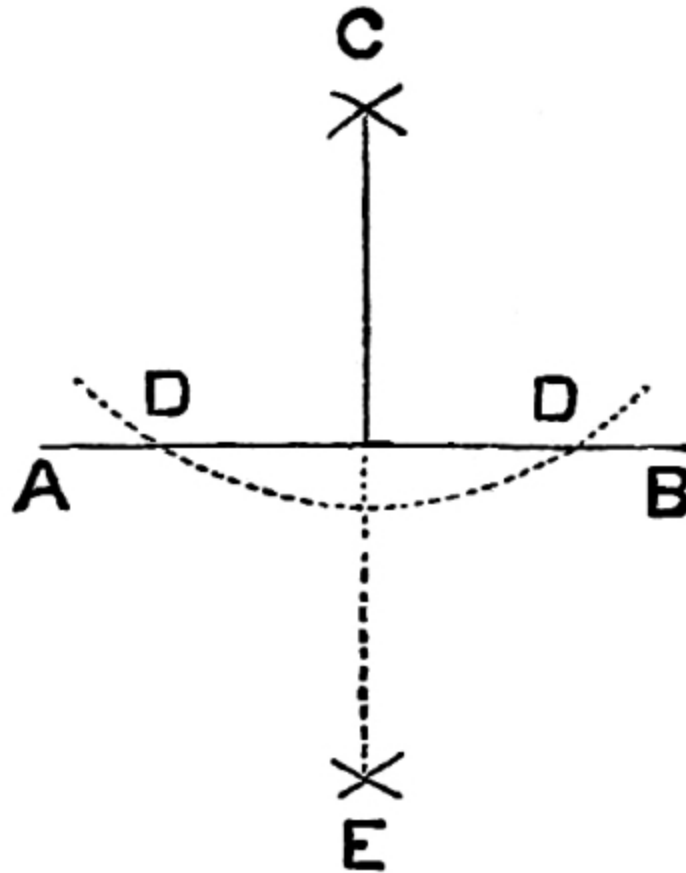
PRACTICAL GEOMETRY.

We will not commence our instructions with the hackneyed “definitions,” but give our readers full credit for the knowledge of what is a *point, a right or straight line, a curved line, parallel lines*—and so forth, and proceed at once to practice.

There are some persons who think that with a drawing-board and square, they can, without fail, make all sorts of horizontal, perpendicular, or parallel lines, and that therefore any geometrical rules for such purpose are to them unnecessary. But, suppose the drawing-board, or the square is absent, or that neither can be had. In such an emergency the want of the following items of knowledge would be severely felt, and, therefore, the acquirement and retention of them is something desirable, and even highly necessary.

PROBLEM I. *To erect a perpendicular on a given right line.*

Fig. 1

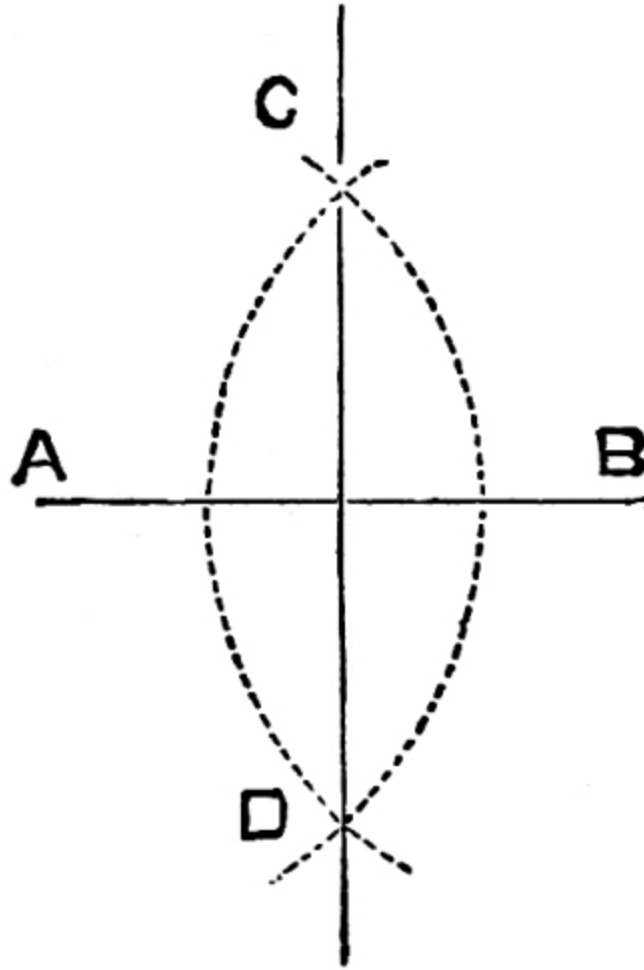


A, B, is the given right line. From the point C, with a radius longer than the perpendicular distance describe the arc, or part of a circle, D, D. And from the points of intersection with the right line A, B, describe arcs cutting each other at C and E. Join C and E, and the perpendicular is obtained on either side of the right line A, B.

PROBLEM II. *To erect a perpendicular at the middle of a right line.*

From the extreme points of the right line A, B, with radii less than the length of the line describe two arcs intersecting each other at C and D, and through the points of their intersection draw the line, which will be perpendicular to the given right line at the middle.

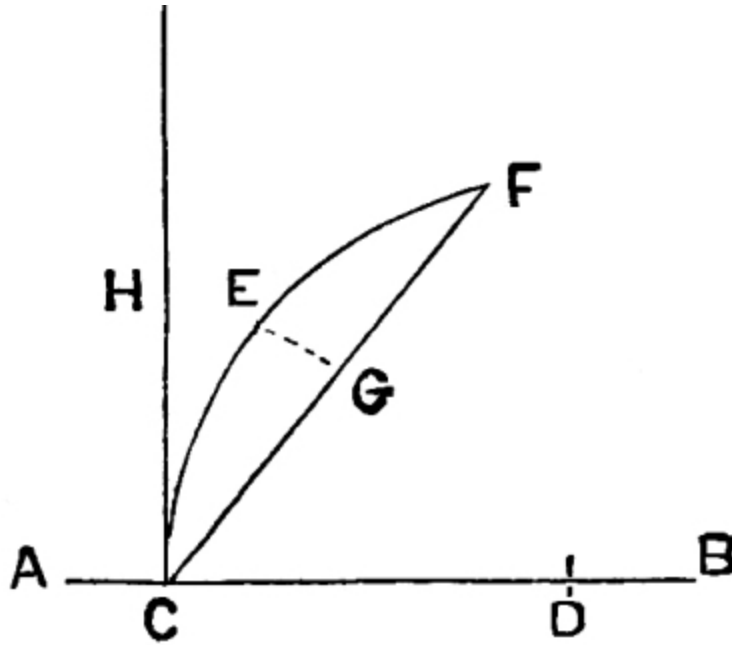
Fig. 2



In this way, too, may any line be divided into two equal parts with facility and exactness.

PROBLEM III. *To erect a perpendicular at or near the end of a given right line.*

Fig. 3



Take any point, D, on the given right line A, B, as a centre, and to the required point C, as a radius, and describe an arc C, E, F. Take a portion of this arc, say E, and make from C, E, equal to E, F. Join F and C. Now with E, C, for a radius, describe the arc G, E, H, and make from E to H equal to from E to G. Then through H from C draw the perpendicular required.

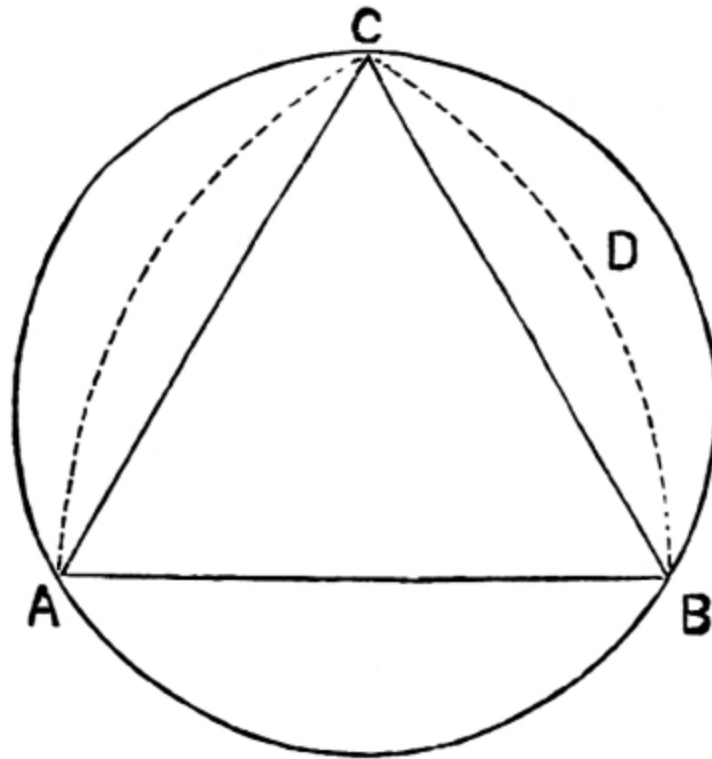
There are other methods of accomplishing this, but we will not introduce them here, as the one now given is sufficient.

We will now proceed to the formation of geometrical figures which enclose space.

That which is bounded by one line is called a *circle*; and a right line dividing it into two equal parts is called its *diameter*; from the centre of which to either end is called the *radius*: and the boundary line is termed the *circumference* from the Latin words *circum*, around, and *fero* to carry. That is: a line carried around. Thus we see an area or space is enclosed by one line. An area may be enclosed by two lines; but one, or both of them, must be curved; as two right lines cannot enclose a space. But three can; and the figure is called a *triangle*.

PROBLEM IV. *In a given circle to construct a Triangle.*

Fig. 4



Take the radius of the circle, and with it mark off six points on the circumference. Take two of these lengths of the radius and join their extreme points A and B, which will be the base. Now take this base as a radius and describe alternately two arcs cutting each other at C. Join A, C, and B, C, and a triangle is formed, whose sides being equal is termed an *equilateral triangle*.

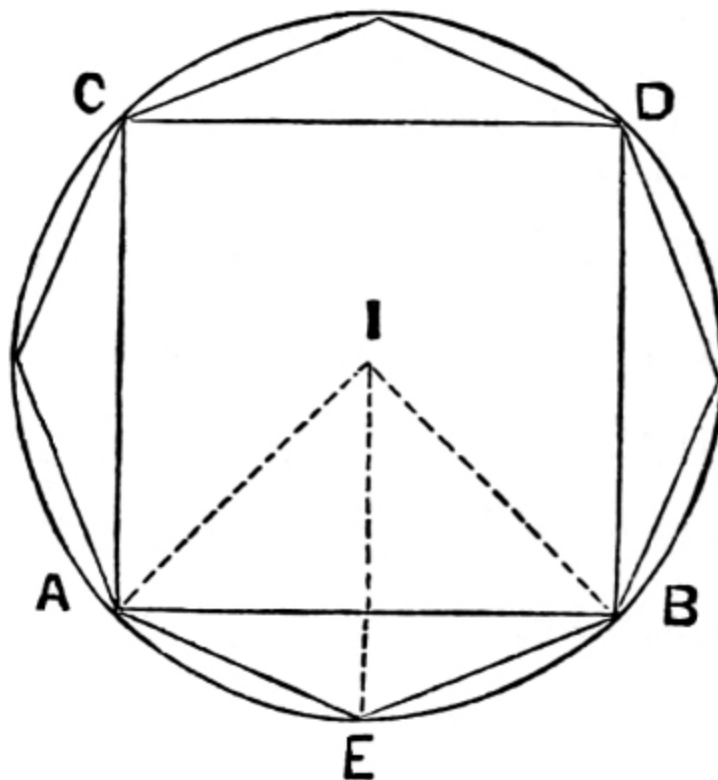
In order to ensure its being upright, erect a perpendicular at the centre, and let the two sides A, C, and B, C, meet that perpendicular where it intersects the circumferences. Or, begin the triangle at this point, and mark off two lengths of the radius, joining the extreme points as before; and do this at each side of the perpendicular; finally connecting the distant extremities of the two sides for a base.

PROBLEM V. *To construct an upright square in a given circle.*

Let fall a perpendicular, I, E, from the centre to the circumference, and with that as a radius and E as a centre, cut the circumference at A, B, C, and

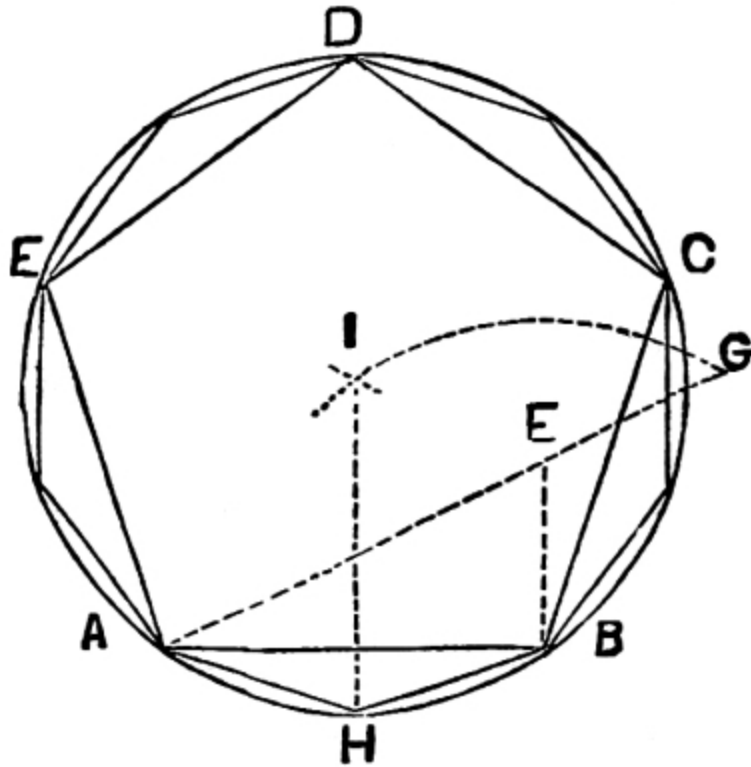
D, and join the points. The four-sided figure called a square is thus formed.

Fig. 5



PROBLEM VI. *On a given right line, A, B, to construct a pentagon, or five-sided figure.*

Fig. 6



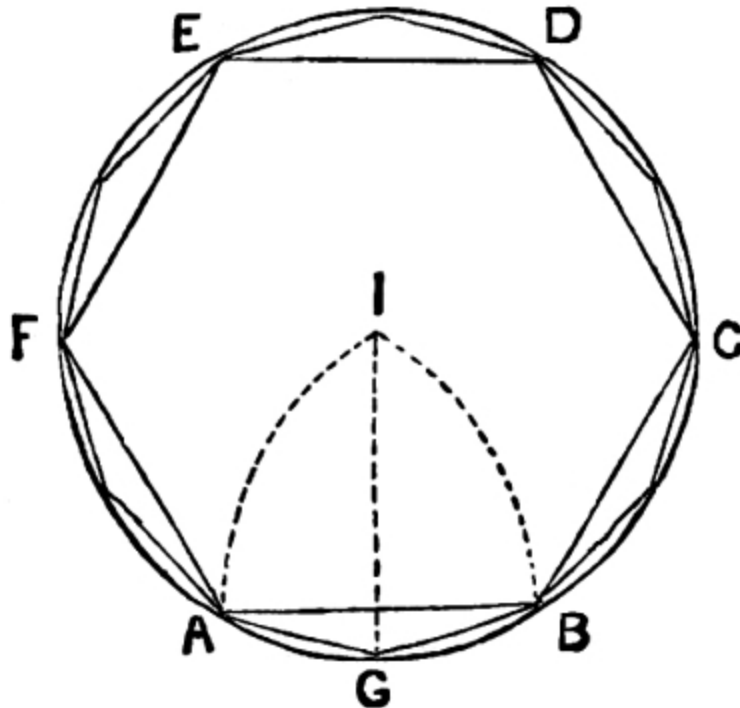
Draw B, F, perpendicular and equal to the half of A, B. Produce A, F, to G, making F, G, equal to F, B. From the points A and B, with the radius B, G, describe arcs cutting each other at I. From I, with the radius I, B, describe a circle. Inscribe the successive chords A, E; E, D; D, C; C, B, which with the base A, B, completes the pentagon.

If the circle be given, and a pentagon to be inscribed in it, the following is as simple as it is practical. From the centre erect a perpendicular, which shall meet the circumference at D. At each side of this point divide the circumference into five equal parts, and connect every two of them from D to E, from E to A, and from D to C, C to B. Now connect A and B and the pentagon is formed.

PROBLEM VII. *On a given line A, B, to construct a hexagon, or six-sided figure.*

Take the length of the radius I, G, and lay it off from F to A, A to B, B to C, C to D, D to E, and E to F.

Fig. 7



PROBLEM VIII. *To form an octagon, or eight-sided figure.*

Refer back to [Fig. 5](#). Draw the radius I, E, till it meets the circumference at E. Join the points E, A, and E, B. Repeat this at each of the four sides, and the octagon is formed.

PROBLEM IX. *To form a decagon, or ten-sided figure.*

Refer to [Fig. 6](#), and proceed as in the preceding problem.

PROBLEM X. *To construct a duo-decagon, or twelve-sided figure.*

Refer to [Fig. 7](#), and duplicate the chords, as already shown.

We do not present 7, 9, or 11 sided figures, because they seldom or ever come into practice. Our object being to give what is useful and not overburden the memory unnecessarily.

The learner should go over and work out each of the foregoing problems several times. In fact, until they are soundly secured in his memory, so that on any emergency he can apply them to a required practice. They are the simplest rudiments, but as practically useful as they are simple. The Architect, the builder, as well as the several trades of carpenter, joiner, carver, stone-cutter, mason, and in fact, all in any way concerned in

the practice of construction will at some time or other wish to recall one of these useful problems. Therefore do we dwell on the necessity for committing them, understandingly, to memory, and likewise the advantage required in being able to draw them neatly and perfectly on paper. In order to do this with satisfaction to one's self, it is desirable that a fine point be constantly maintained on the pencil, and that uniform nicety be preserved with the curved lines, as well as the right or straight lines. For nothing looks worse than undue thickness in the one or the other. All should be alike.

In theoretical geometry a line, whether right or curved, is but imaginary, not having any thickness whatever, and therefore no palpable existence. In practical geometry the line must be visible, but ought to be so uniformly fine as to occupy scarcely any perceptible thickness. And herein lies the greatest beauty in geometrical draughting. By strict attention to this apparently trifling matter, its advantages will show wherever minute angles occur. They will be clear and distinct, and always satisfactory.

The learner should keep his first attempts, however coarse, for they will by comparison hereafter, show the advance he has made. Nor should he be content to "let well enough alone." There is no "well enough" in drawing. It is a progressive science, and the true artist never believes he has done his best. Go as near to perfection as you can, and do not turn aside from, or step over obstacles to reach the end you have in view. Whatever you have neglected in early study will surely haunt you through after years, and trouble you when you can least bear the annoyance.

We now conclude this primary lesson, hoping that our learners may profit by the hints we have thrown out, and will thoroughly prepare themselves for the advance in our next.

THE first brick house in Iowa was built by Judge Rerer, of Burlington, in 1839.

VINES FOR THE DECORATION OF COTTAGES.

THE GROUND NUT VINE.

A tourist riding a few miles in almost any New England city, would hardly fail to notice that a large number of the rural residences display a profusion of architectural embellishment, without wearing a cheerful, home-like look. He would pass cottage after cottage ornamented with slender porticoes, fanciful verandas, sculptured gables and deep bay windows, but situated in a pen-like looking enclosure, and surrounded with fixtures, dark and dismal; and with arbor vitæ hedges whose yellow cast clearly indicated that they had been planted in ungenial soil. In each narrow yard he would notice flower beds, containing many unhealthy looking plants, and most of these beds would exhibit the same arrangement and the same multifarious specimens of the odds and ends of Nature for miles. He would remark concerning these suburban seats that they were *pretty*; he would hardly say beautiful, certainly not charming. They were not satisfying to the eye—they were designed to impart an expression of exquisite rurality but failed. As the same tourist passed by some old-fashioned farm house, with its broad green lawns in front, shaded with green old elms; as he noticed the wood colored porch covered with luxuriant woodbine, the dove-cote with its glittering birds, the dark orchards beyond the yard, the pond in the meadow overhung with willows; or, as he descried some inexpensive cottage, removed from the road and half hidden from view by graceful arbors and vigorous native trees, he would ride slowly and express his satisfaction at each of these scenes of rural taste and beauty.

It is not the richness of art that gives to English cottages their picturesqueness and poetic expression, but the beauty of the grounds that surround them, and the vines that adorn them. It is not the fantastic gables, nor the latticed windows that so captivate the eye of the traveller, but the tasteful foliage that drapes them, and the lustrous vines that embower them. Denude these cottages of these embellishments, and many of them would

appear as uninviting to the eye as the mouldering tower without the classic ivy.

Louis XIV had his Versailles, and his elegant queen her embowered Triannon; but the simple charms of Triannon proved more inviting to the cultivated minds of the court, than the gorgeous pile and artificial gardens at Versailles.

We devote too much time to the cultivation of exotics, and too lightly value the vines and shrubs of our native soil. Again, we sacrifice rich foliage that lasts for a season, to gaudy flowers that last only for a brief period. The double prairie rose is a very delightful sight—for a single week—and during the remaining season it is a miserable brier, commonly wormy and lousy. Yet the prairie rose is in common use as an ornament for the veranda, while the jessamine, the woodbine, the wisteria and the luxuriant honeysuckle are, put in less conspicuous places, or their cultivation wholly neglected.

It may be cited as an evidence of improving taste in the rural art, that rustic work, which imparts to a place an expression of delightful rurality, is taking the place of images, porcelain vases, etc., that long have been conspicuous objects in almost every parterre. The perfection of beauty to which this work may be carried has been admirably illustrated in Central Park, New York City, and widely copied by gentlemen of taste. Few objects are more pleasing than rustic arbors or even rustic urns over-running with foliage.

Among the most pleasing vines for embellishment of rural seats are the honeysuckle (*Lonicera japonica* and the trumpet vine), the woodbine, the jessamine and the American ivy. For adorning stone work, the English ivy is very rich, though it grows imperfectly in our Northern latitudes.

The woodbine forms a massive drapery for a cottage porch. It has a rich marine hue in summer, and it is very richly tinted in autumn after the early frosts. The Japan honeysuckle is deliciously fragrant, and it retains its dark lustrous foliage until mid-winter. Unlike many climbers this honeysuckle, together with the trumpet vine, is not liable to be infested with insects. The feathery *clematis*, known also by the names of the *virgin's bower* and the *traveller's joy*, is a pretty creeper for walls and fences; and the common hop

vine may be made to add beauty to the dove-cote and the martin boxes, when these are placed after the old English manner, upon poles.

The American ivy is one of the most prolific of foliage vines. The leaves when they are young are of a delicate pea-green color, but they become dark and lustrous as the season advances. They are very gorgeous after the early autumn frosts, displaying the richest tints of orange and vermillion. The ivy forms a sort of net-work for old crumbling walls, and it is indigenous to stormy places.

There is a slender vine very common in the Eastern States that is seldom used for ornamental purposes, to which we would especially invite the attention of the florists. It is called the ground nut, (*Apios tuberosa*.) Its foliage is dark, thick, and very graceful. The flowers are remarkable. They are dark purple in color and present a peculiar waxy appearance, in dense predunculate, axillary racemes. Their odor is wonderfully sweet, and it is so powerful and inexhaustive as to fill perpetually the air. The vine entwines itself among low bushes in its native state. A florist of our acquaintance supplemented the charms of her trellises of roses by entwining these vines among the branches. Her rooms were filled with fragrance whenever the windows were thrown open during the whole of the hot season. The flowers of the ground nut vine last for a very long period. Remember this vine in your summer rambles.—*Working Farmer*.

CLEAN THE CELLAR.—The *Boston Journal of Chemistry* says: “Diphtheria, typhoid, and scarlet fevers, and many other most serious illnesses, have their origin in cellars both in city and country; and we can do our readers no greater service than to urge them to see that at all times they are in a dry, sweet, wholesome condition. Why should farmers and farmer’s families, living in the country away from the pestilential vapors of the cities, be so subject to attacks of malignant diseases? There is a reason for it, and we can point it out. They arise from the indifference manifested to the observance of hygienic rules and the violation of sanitary laws. Cleanliness is essential to health, and it is just as necessary in the country as in the city. A family living over a foul cellar is more liable to be poisoned and afflicted with illness than a city family living in its polluted atmosphere, but without cellar or basement filled with fermenting roots and fruits. There is far more sickness in the country among husbandmen than there ought to be. With plenty of pure air, water, and exercise, the evil imp Disease ought to be kept

at bay, and he would be better if an observance of certain hygienic conditions were maintained. Bad conditioned cellars, small, close sleeping rooms, stoves—these are all agents of evil, and are fast making the homes of farmers almost as unhealthy as those of the dwellers in cities. Are not these suggestions worthy of consideration?”

ON THE ART OF GARDENING.

BY THOMAS HOPE.

What was, in the earlier times, the origin of the garden? The wish that certain esculent plants and fruits, which in the waste field and the wide forest are scattered at great distances, in small quantities, intermixed with useless vegetables and fruits, precarious in their appearance, and stunted in their growth, difficult to collect, and scarce worth the gathering, might in a nearer, a smaller, and a more accessible spot, be better secured, more abundantly produced, kept clearer of the noxious herbs and weeds which destroy their nutriment and impede their growth. This was, in its origin, the sole object of the entire garden; this, to the present hour, continues to be the principal purpose of that essential portion of the garden, devoted to the uses of the kitchen and the table.

In these parts of the garden then, which are destined immediately for the gratification, not of the eye, but merely of the palate, it is only in proportion as we more fully deviate from the desultory and confused dispositions of simple nature—firstly, by separating the different species of esculent plants, not only from their useless neighbors, but from each other; and secondly, by confining the vegetables thus classed in those symmetric and measured compartments, which enable us with greater ease to discover, to approach, and to improve each different species in the precise way, most congenial to its peculiar requisites, that we more fully attain that first of intellectual beauties, which, in every production, whether of nature or of art, resides in the exact correspondence between the end we propose and the means we employ. Nay, if it be true that contrast and variety of colors and of forms are amongst the most essential ingredients of visible beauty, we may say that even this species of sensible charm is greatly increased in the aspect of a country by the opposition to the more widely diffused, but more vague shades and outlines of the unsymmetrical surrounding landscape, offered by the more vivid hues and more distinct forms of the gay Mosaic work of nicely classed and symmetrized vegetables which clothe these select spots.

Even where the general unadorned scenery is as bold and majestic as in Switzerland, or as rich and luxuriant as in Sicily, the eye with rapture beholds the variety, and enjoys the relief from the vaster and sublimer features of rude Nature, offered by the professed art of a neat little patch of ground, whether field, orchard, or garden, symmetrically distributed. It looks like a small but rich gem—a topaz, an emerald, or a ruby, sparkling amidst vast heaps of ruder ore; or rather like a rich carpet, spread out over a corner of the valley. It appears thus incontrovertible, that in that part at least of the garden which is immediately intended for utility, we incidentally produce not only greater intellectual, but greater visible beauty, by not confining ourselves to the desultory forms of unguided Nature, but by admitting the more symmetric outlines of avowed art, and it therefore only remains to be inquired, whether in that other and different part of the artificial grounds, in later times added to the former, which is directly intended for beauty, and which we therefore call the pleasure-grounds, we shall really produce more beauty, intellectual or visible, or, in other words, more pleasure to the mind or eye, by only employing the powers of art in a covert and unavowed way; in still only preserving the closest resemblance to the interminable and irregular forms of mere nature, or by exhibiting her additional resources in a more open and avowed manner; in contrasting these more indeterminate and desultory features of pure nature, with some of those more determinate and compassed outlines, which, indeed, on a small scale, are already found in many of the spontaneous productions of Nature herself; but which on a more extended plan, are only displayed in the works of art. I say, more pleasures to the mind or eye; for the portion of the garden here alluded to, no less than the one before mentioned professes itself to be a piece of ground wrested from Nature's dominion by the hand of man, for purposes to which Nature alone was inadequate; and thence contending that there is the least necessity or propriety in rendering this district, appropriated by art, a fac-simile of pure Nature, independent of any consideration of superior beauty which this imitation may offer to the eye or mind, and merely because, to form a garden, we use materials supplied by Nature—such as air, water, earth, and vegetables, would be absurd in the extreme. As well might we contend, that every house, built of stone should resemble a cavern, and every coat made of wool, a sheepskin. Every production of human industry whatsoever, must, if we trace it to its origin, arise out of one or more definite ingredients of pure nature; and unless,

therefore, by the same rule, every production of human industry whatsoever be obliged everlastingly to continue wearing the less regular forms of those peculiar objects of nature, out of which it is wrought, we cannot with more justice arraign gardens in their capacity as aggregates of mere natural substances and productions, for assuming the artificial forms of a terrace or a *jet-d'eau*, an avenue or a *quincunx*, than we can condemn opera-dancers and figurantes, in their capacity of compounds of natural limbs and features, for exhibiting the artificial movements of the minuet and the gavot, the entrechat and the pas-grave.

If, then, the strict resemblance to the desultory forms of rude nature be not indispensably requisite in the artificial scenery of pleasure-grounds, on account of any invariable reasons of propriety or consistency, inherent in the very essence of such grounds, this resemblance of studious art to wild nature, in the gardens that adorn our habitations, can only be more eligible on account of some superior pleasure which it gives the eye and mind, either in consequence of certain general circumstances connected with the very nature of all imitation, or only in consequence of certain more restricted effects, solely and exclusively produced by this peculiar species of imitation; namely, of natural landscapes through artificial grounds.

Now, with regard to the former of these two considerations, I allow that a faithful imitation, even of a deformed original, is capable of affording great intellectual pleasure to the beholder, provided that imitation, like that displayed in painting and sculpture, be produced through dint of materials, or tools so different from those of which is composed the original imitated, as to evince in the imitator extraordinary ingenuity and powers; but the imitation of a natural landscape, through means of the very ingredients of all natural scenery; namely, air, earth, trees, and water, (and which imitation will in general offer greater truth in proportion as it is attained through greater neglect,) cannot possess that merit which consists in the overcoming of difficulties and the display of genius; unless, indeed, it be an imitation of such a species of wild scenery as is totally foreign to the genius of the locality in which it is produced; unless it consists in substituting mountains to plains, waterfalls to puddles, and precipices to flats; and in that case, on the contrary, the attempt at imitation will become so arduous as to threaten terminating in a total failure, by only offering, instead of a sublime and improved resemblance, a most paltry and mean caricature. Since, then, in a

garden, the imitation of the less symmetric arrangements of rude nature can afford little or no peculiar gratification to the mind in their sole capacity as imitations, the question becomes restricted within a very narrow compass; and all that remains to be inquired into is, whether, in that garden, the exclusive admission of mere unsymmetric forms of simple nature, or their mixture with a certain proportion of the more symmetric forms of professed art, will give more intense and more varied pleasure to the eye? And, when thus stated, I should think the question would be nearly answered in the same way by every unprejudiced person. I should think it would be denied by none, that if, on the one hand, the most irregular habitation, still, through the very nature of its construction and purposes, must ever necessarily remain most obviously symmetric and formal; if not in its whole, at least in its various details, of doors, windows, steps, entablatures, etc., and if, on the other hand, as I take it, all beauty consists in that contrast, that variety, that distinctness of each of the different component parts of a whole, from the remaining parts, which renders each individually a relief to the remainder, combined with that harmony, that union of each of these different component parts of that whole with the remaining parts, which renders each a support to the remainder, and enables the eye and mind to glide over and compass the whole with rapidity and with ease, fewer striking features of beauty will be found in a garden, where, from the very threshold of the still ever symmetric mansion, one is launched in the most abrupt manner, into a scene wholly composed of the most unsymmetric and desultory forms of mere nature, totally out of character with those of that mansion; and where the same species of irregular and indeterminate forms, already prevailing at the very centre, extend, without break or relief, to the utmost boundaries of the grounds, than will be presented in another garden, where the cluster of highly-adorned and sheltered apartments that form the mansion, in the first instance, shoot out, as it were, into certain more or less extended ramifications of arcades, porticoes, terraces, parterres, treillages, avenues, and other such still splendid embellishments of art, calculated by their architectural and measured forms, at once to offer a striking and varied contrast with, and a dignified and comfortable transition to, the more undulating and rural features of the more extended, more distant, and more exposed boundaries; before, in the second instance, through a still further link, a still further continuance of this same gradation of hues and forms, these limits of the private domain are again made in their turn, by means of

their less artificial and more desultory appearance, to blend equally harmoniously on the other side, with the still ruder outlines of the property of the public at large.

No doubt, that, among the very wildest scenes of unappropriated nature, there are some so grand, so magnificent, that no art can vie with, or can enhance their effect. Of this description are the towering rock, the tremendous precipice, the roaring cataract, even the dark, gloomy, impenetrable forest. Of such, let us take great care not to destroy, or to diminish the grandeur by paltry conceits or contrivances of art. But even these are such features as, from certain conditions unavoidably attendant on them, we would not wish to have permanently under our eyes and windows; or even if we wished it, could not transport within the narrow precincts which immediately surround the mansion. A gentleman's country residence, situated in the way it ought to be, for health, for convenience, and for cheerfulness, can only have room in its immediate vicinity for the more concentrated beauties of art. In this narrow circle, if we wish for variety, for contrast, and for brokenness of levels, we can only seek it in arcades and in terraces, in steps, balustrades, regular slopes, parapets, and such like; we cannot find space for the rock and the precipice. Here, if we admire the fleeting motion, the brilliant transparency, the soothing murmur, the delightful coolness of the crystal stream, we must force it up in an erect *jet-d'eau*, or hurl it down in an abrupt cascade; we cannot admit so near us the winding torrent, dashed at wide intervals from rock to rock. Here, if we desire to collect the elegant forms, vivid colors, and varied fragrance of the choicest shrubs and plants, whether exotics, or only natives, oranges, magnolias, and rhododendrons, or roses, and lilies, and hyacinths; we still must confine them in the boxes, the pots, or the beds of some sort of parterre; we cannot give them the appearance of spontaneously growing from amongst weeds and briers. Here, in fine, if we have a mind to secure the cool shade and the convenient shelter of lofty trees, we can only plant an avenue, we cannot form a forest. And for that, since we admire, even to an excess, symmetry of lines and disposition in that production of art called a house, we should abhor these attributes in the same excess in that other avowed production of art, the immediate appendage of the former, and consequently the sharer in its purposes and character, namely, the garden, I do not understand. There is between the various divisions of the house and those of the grounds, this difference, that the first are more intended for

repose, and the latter for exercise; that the first are under cover, and the latter exposed. The difference should make a corresponding difference in the nature of the materials, and in the size and delicacy of the forms; but why it should occasion on the one side an unqualified admission, and on the other, as unqualified an exclusion of those attributes of symmetry and correspondence of parts which may be equally produced in coarser as in finer materials, on a vaster as on a smaller scale, I cannot conceive. The outside of the house is exposed to the elements as well as the grounds; and why, while columns are thought invariably to look well at regular distances, trees should be thought invariably to look ill in regular rows, is what I cannot comprehend. Assuredly the difference is as great between the eruptions of Etna, or of any other volcano, and artificial fire-works, as it is between the falls of the Niagara or of any other river, and artificial water-works. Why, then, while we gaze with admiration on a rocket, should we behold with disgust a *jet-d'eau*? And why, while we are delighted with a rain of fiery sparks, should we be displeased with a shower of liquid diamonds, issuing from a beautiful vase, and again collected in as exquisite a basin? If the place be appropriate, if the hues be vivid, if the outlines be elegant, if the objects be varied and contrasted, in the name of wonder, how should, out of all these partial elements of positive, unmixed beauty, arise a whole positively ugly? No, there can only arise a whole as beautiful as the parts; and so, those travellers who have not allowed any narrow and exclusive theories to check or destroy their spontaneous feelings, must own they have thought many of the suspended gardens within Genoa, and of the splendid villas about Rome; so they have thought those striking oppositions of the rarest marbles to the richest verdure; those mixtures of statues, and vases, and balustrades, with cypresses, and pinasters, and bays; those distant hills seen through the converging lines of lengthened colonnades; those ranges of aloes and cactuses growing out of vases of granite and of porphyry, scarce more symmetric by art, than these plants are by nature; and, finally, all those other endless contrasts of regular and irregular forms, everywhere, each individually increasing its own charms, through their contrast with those of the other, exhibited in the countries, which we consider as the earliest schools, where beauty became an object of sedulous study.

But the truth is, that in our remoter climes, we carry every theory into the extremes. Once, that very symmetry and correspondence of parts of

which a certain proportion ever has, to all refined ages and nations, ancient and modern, appeared a requisite feature of the more dressy and finished parts of the pleasure garden, prevailed in all English villas with so little selection, and at the same time, in such indiscreet profusion, as not only rendered the different parts insipid and monotonous with respect to each other, but the whole mass a most formal, unharmonious blotch with regard to the surrounding country. Surfeited at last with symmetry carried to excess, we have suddenly leaped into the other extreme. Dreading the faintest trace of the ancient regularity of outline as much as we dread the phantoms of those we once most loved, we have made our country residences look dropped from the clouds, in spots most unfitted to receive them; and, at the expense, not only of all beauty, but of all comfort, we have made the grounds appear as much out of harmony, viewed in one direction with the mansion, as they formerly were viewed in the opposite direction with the country at large. Through the total exclusion of all the variety, the relief, the sharpness, which, straight or spherical, or angular, or other determinate lines and forms might have given to unsymmetric and serpentine forms and surfaces, we have, without at all diminishing the appearance of art, (which in a garden can never be totally eradicated,) only succeeded in rendering that art of the most tame and monotonous description; like that languid and formal blank verse, which is equally divested of the force of poetry and the facility of prose. Nature, who, in her larger productions, is content with exhibiting the more vague beauties that derive from mere variety and play of hues and forms; Nature herself, in her smaller and more elaborate, and if I may so call them, choicer bits of every different reign, superadds those features of regular symmetry of colors and shapes, which not only form a more striking contrast with the more desultory modifications of her huger masses, but intrinsically in a smaller space, produces a greater effect than the former can display. Examine the radii of the snow-spangle, the facettes of the crystal, the petals of the flower, the capsules of the seed, the wings, the antennæ, the rings, the stigmata of the insect and the butter-fly; nay, even in man and beast, the features of the face, and the configuration of the eye, and we shall find in all these more minute, more finished, and more central productions of the mineral, the vegetable, and the animal kingdoms, reigns the nicest symmetry of outline and correspondence of parts. And if art, which can only be founded upon, only spring out of nature; if art, I say, should ever

only be considered as the further development of nature's own principles, the complement of nature's own designs, assuredly we best obey the views of nature, and best understand the purposes of art, when, leaving total irregularity to the more extended, more distant, and more neglected recesses of the park, we give some degree of symmetry to the smaller and nearer, and more studied divisions of the pleasure-ground. This principle of proportioning the regularity of the objects to their extent, the Greeks well understood. While in the Medici Venus the attitude of the body only displays the unsymmetric elegance of simple nature, the hair presents all the symmetry of arrangement of the most studious art; and unless this principle also become familiar among us there is great danger that unable to make the grounds harmonize with the mansion, we attempt to harmonize the mansion with the grounds, by converting that mansion itself into a den or a quarry.

REMARKS ON FIRE-PROOF CONSTRUCTION.

A PAPER READ BEFORE THE NEW YORK CHAPTER OF THE
AMERICAN INSTITUTE OF ARCHITECTS, APRIL 6TH, 1869.

BY P. B. WIGHT, F. A. I. A.

MR. PRESIDENT AND GENTLEMEN:—A distinguished member of this body not long since remarked that a fire-proof building was easily defined: “It is a building which cannot burn, and which contains nothing that will burn.” Admitting the definition, I do not propose to dispute with the gentleman, neither do I intend to enter into an elaborate and scientific investigation of the subject; to do so would be to essay a task far beyond my powers, and one which might result in stultifying myself and wearying you. The best I can do is to collect some of the scattered results of thought and observation, into what I trust you will consider to be but a rambling dissertation upon a subject which is of great interest to all of us. It is, therefore, less with the desire to display any erudition, than to introduce the subject, and call forth the views of those assembled here, that I have chosen to address you some remarks on fire-proof buildings. In so doing it is possible that I may enter the field of criticism, and may comment upon the works of some who are here present; but whatever I may say in that direction, allow me to assure you, will be said with justice and candor, and an endeavor to follow Matthew Arnold’s definition of criticism—to find the best ideas in everything. I will look to those whose experience has been more extended than mine, for a continuation of the discussion of what I may only hint at.

It is very seldom that any building is required for such use that only non-combustible material shall be placed in it; but it is still a fact that fire-proof buildings are often called for, and are needed, wherein large amounts of combustible materials are to be placed.^[A] To supply such a demand, is one of the most important problems offered to the architect for solution. Of such buildings, are storage warehouses, and stores or shops, wholesale and

retail, as well as buildings for certain kinds of manufacturing processes, such as sugar-houses and carriage or furniture shops.

Having devised a building of non-combustible material throughout, the question which next arises is how to keep a conflagration in one part from extending to all the contents of the building. It seems to me, that in buildings for such purposes, the idea of making them only partially fire-proof is not to be considered for a moment, unless, perhaps, the material contained is so highly inflammable that it would destroy the material of the building, even if it is divided into fire-proof compartments, in which case it seems to be folly to go to the expense of fire-proof materials at all. When you know that no part of your building can burn of itself it is evident that every atom of it will offer some resistance to the enemy confined within. I believe, too, that it is impossible to smother or choke a fire once commenced, by the use of closed compartments. Accident or carelessness may leave some openings which will facilitate a draft in some unforeseen way. And even supposing that you have shut in your fire by some arrangement of closed compartments, can you give your compartments less air than a charcoal pit? Close it as much as you will, your confined goods, if the barriers are not forced by the immense power generated by the heat, will at last be reduced to charcoal; for you cannot open a door or window upon such a smouldering fire, but that it will instantly burst into flames. Ships have been brought to port with smouldering fires under their closed hatches, which have been in existence for weeks at a time, while but few have been eventually saved under such circumstances, except by scuttling. Such conditions do not exist with regard to buildings; in them there is not the risk of human lives, which may be saved on shipboard only by closing down the hatches, and scuttling is obviously out of the question.

Store-houses are the only class of buildings which admit of division into airtight compartments, and there is a practical objection to them in even buildings of this class; but few kinds of goods can be preserved without good ventilation. It seems, therefore, that the compartments should be open and accessible from without, but carefully divided from each other. If so, they afford good facilities to those employed in extinguishing fires; and I think that in a building thus arranged, there would be a more reasonable chance of a portion of its goods being saved.

The division of buildings into horizontal compartments, rather than vertical ones, is so much more desirable, where land is expensive, that inventors have almost exhausted their ingenuity in devising thoroughly fire-proof floors. It is obvious, however, that the division of a building by vertical fire-proof partitions, is a matter so easy of accomplishment, that it is questionable whether the horizontal division, so beset with practical difficulties, so expensive, and withal so much less to be depended upon, even when the best systems of construction are used, is ever economical, even where ground is expensive. I even question whether it is of any use to build iron floors, or floors with iron supports, for buildings to contain goods; brick piers and groined arches are alone reliable. If you divide horizontally you must have stairways within and windows on the exterior, both of which welcome the ascending flames. You may enclose your staircase in a fire-proof enclosure, and you may put the heaviest iron shutters on your windows, but you must have doors through which to gain access from your stairways, and you must open your shutters when you want light. There is a contingency that these traps may be set when the enemy comes, and then all your expensive floors represent so much wasted capital.

As yet, I believe, that no buildings in this vicinity, built purely for storage purposes, have been constructed entirely of fire-proof materials, except the St. John's Depot of the Hudson River Railroad Company. I am not aware that any attempt has been made in these buildings to stop a conflagration among the goods on storage either by horizontal or vertical compartments. The floors, to be sure, are of iron and brick, non-combustible, but with hoistways; and it is not difficult to conjecture, even supposing that all horizontal openings and iron shutters were closed, what would be the result of a fire raging on one of those floors, hundreds of feet in expanse.

Several fires occurring recently in the Brooklyn warehouses have warned their owners to take extra precautions, even though none of these warehouses are fire-proof, if I am rightly informed. One of the best is known as the Pierrepont Stores, near the Wall Street Ferry, and the arrangement of them is well worthy of notice. These are about three hundred feet in length, and are divided into six compartments by fire-proof party walls; the width of each compartment is consequently about fifty feet,

and the length about two hundred feet. The floors are of wood, and it would have been useless to make them of iron and brick; for the goods taken in them are mainly sugars, and it would be folly to attempt to arrest a fire of such combustible material in its ascending course, by any practicable device. But what is most interesting in these buildings is that each is fortified against its neighbor. Recently the party walls were carried up about six feet above the roofs, and were pierced with embrasures, through which firemen can play from the roof of one building upon the flames of another, with perfect safety to themselves. Here is an instance wherein capital would have been wasted on the expensive materials required for fire-proof floors.

It is the duty of the architect, as I conceive it, to guide the capitalist in coming to a decision on such points. If he devises economical methods, his commission is lessened, but thereby so much more capital remains unemployed, but ready for investment in other enterprises. It would be foreign to my subject to enlarge upon this point, and show how much more it is to the interest of the architect to study reasonable economy in his works, especially buildings for business purposes; but I will let the suggestion stand for what it is worth. Perhaps a knowledge of the fact that most members of our profession agree with me in this opinion would go far toward disarming the misgivings of many a client upon the question of commissions.

Buildings for manufacturing purposes next demand attention. Some time since a manufacturer and contractor for iron work remarked to me, that if some one would only put up a large fire-proof building, with good steam power, to be rented out for manufacturing purposes, his fortune would easily be made. I have often thought of the suggestion, and wondered why it had not been acted upon. He said that at that time it would be impossible to hire a fire-proof shop or room, with power, in this city. Now, there are many occupations requiring delicate, and not easily replaced machinery, or in which are involved elaborate experiments, running for long periods—the derangement of which could not be recompensed by any amount of insurance—for which a fire-proof building would be almost invaluable. The saving of insurance on such a building and its contents would be greater than the interest on the extra cost of fire-proof floors, and would enable the owner to rent his rooms at a lower rate—in proportion to the equivalent given—than could the owners of buildings with wooden floors. The extra

cost of fire-proof construction in a manufacturing building is small when compared with that of a bank or public building. The walls and ceilings require neither lath nor furring, and the floors may be of flags or slate, bedded on the brick arches, or what is better, plates of cast-iron bolted to the beams—which will presently be described. All inside finish may be discarded, and iron doors, of No. 16 iron, with light wrought-iron frames, hung to stone templates in the jambs, are the only coverings required for the openings.

Such fire-proof buildings as have been erected for manufacturing purposes have been specially designed for single occupants. The most perfect and the earliest that I know of is a building erected on Vestry street, about ten years since, for the Grocers' Sugar Refining Company. This building, as far as its material is concerned, is absolutely fire-proof. It is most remarkable for its floors, which are made of plates of boiler-iron, riveted together and secured to the beams in large sheets. This is the most simple system of floor construction I have ever seen, and has many advantages. But I have not seen the building in use, and do not know how the floors answer the ends for which they are intended.

Some of the new buildings for the various gas works in this city are fire-proof. The best are those of the Metropolitan Company, at the foot of Forty-second street, North river. But they are at best, only sheds—brick walls, with iron shutters and roofs. Large, open, and well ventilated, they serve their purposes well; but they can hardly be called architecture.

The most extensive attempt to build a fire-proof building for manufacturing purposes was the enterprise of Harper & Brothers. This was one of the pioneer buildings of the new dispensation. The Harper girder is well known; it is an ornamented cast-iron beam, with a tie rod, and was the father of the truss beam, now so extensively used for supporting the rear walls of stores. It has been succeeded by the built-up beam, now generally used for girders, and the double rolled beam. It was eminently a constructive beam, using iron according to its best properties, cast-iron for compression and wrought-iron for tension. I doubt not that it will some day be again used where girders are required. The built-up beam was invented for the restorer of the "pure" styles, who think that furring strips, laths, plaster and a modicum of run moulding, not to forget "a neat panel on the soffit," to be a good substitute for the honest lintel of the Greeks, and more

artistic than the constructive beam which James L. Jackson & Bro. designed and executed for the Harpers. When men are no longer ashamed to display good iron construction, and bend their artistic conceptions to their constructive skill, we may hope to see something like the Harper beam revived, and decorated in a manner befitting its use. But I fear that this will be done when a more rational generation than our own holds the sway. But to return. In Harper's building, as in the Cooper building, the deck beam was used for the floors, and brick arches, such as those now in use, were employed. The deck beam has also gone out of use. When first employed, iron beams were not made for houses, but for ships. The I beam, has replaced the deck beam for the former purpose. And in this connection, I would suggest an inquiry into the practicability of using the deck beam inverted. It has always seemed to me that the broad flange would best sustain compression, and that the roll, having the form of a round bar, would best resist tension. The matter of the bearings is easily remedied by a cast-iron shoe on each end of the beam and bolted to it. This shoe, with a broad foot, would answer the purpose both of template and anchor, and if made to project from the wall and assume an ornamental shape, might become a visible and constructive bracket. The deck beam inverted would evidently present the best appearance from below in cases where the flooring is placed on top of the beams—the various methods of doing which I propose to discuss further on. Should the deck beam come again into use, it might be made of more ornamental form without detriment to its strength. The bottom roll or flange could be moulded in various ways.

But, except in so far as the floors are concerned, the Messrs. Harper's building is far from being fire-proof. There is much wood-work in its inside finish, and the contents being of a highly inflammable nature, I fear that fire would have its own way in that building unless early checked.

Besides these buildings two partially fire-proof publishing houses have been built; the Times Building and the Ledger Building; but there is nothing in either that it is pertinent to my inquiry to mention;—they are manufacturing buildings in the same sense that the Harper's Building is, but the former might as well come within the class of office buildings.

The fact of the American Bank Note Company having taken quarters in the Mutual Life Insurance Building, upon their expulsion from the Custom House, illustrates what my friend mentioned about the demand for buildings

for delicate and elaborate processes, such as the art of bank note engraving, and goes to show that such branches of business are obliged to settle in buildings erected for other purposes. The work of a bank note company is in some respects a heavy manufacturing business, which any one will believe who examines the powerful boilers and engines in the cellar of the Mutual Insurance Building; but it is also a delicate artistic business, requiring steady floors, good light, and absolute safety from fire, to the valuable materials used and kept in it, which not money alone could replace.

From the Bank Note Company we come next to the Assay office whose risks are similar. I am informed that it is absolutely fire-proof, but I have had no occasion to visit it.

Of Banks and Insurance Buildings we certainly have a large number which are to all intents fire-proof, though but few are thoroughly so. It is generally admitted that such buildings are not in danger from their contents, and to this belief may be ascribed the fact that we already have so many of this class. The Continental Bank, the American Exchange Bank, the Mutual Life Insurance Company's building, the Park Bank, and the City Bank building, recently remodeled, are absolutely fire-proof. Nothing less than a bonfire of all the furniture, books, and papers that could be collected together in any one room of any of these buildings would endanger its destruction. They are safe from any ordinary casualty. But in all the rest there is enough wood-work to make the word "fire-proof," as applied to them, of very doubtful significance. To show what a practical eye the Insurance Companies have, let me say that in nearly all the so-called fire-proof bank buildings the rates of insurance are as high as in ordinary business buildings. The rates are unusually high in the building which I happen to occupy, on account of a well hole in the centre which is trimmed with wood, and would carry a fire through the whole building in an instant. What I might say in relation to buildings of this class will be comprised in some practical suggestions upon fire-proof buildings generally. Let us then look for a few moments into the matter of constructive details.

And, firstly, how shall floors be constructed? Before the "iron period," when our Washington Capitol, our City Hall, our old Exchange and Custom House were built, the Roman Mediæval vaults only, were used—either of stone or of brick plastered. When the width of a room was too great for one span, granite columns or brick piers were used, as in our old Exchange, now

the Custom house. The floors above the vaults were leveled up and paved with flags or marble tiles. As far as grace, strength and absolute relief from the dangers of fire were concerned, this was a perfect system. But now space is demanded; there must be no more heavy piers and no great thickness of floors. We are therefore forced to use a material which, though not combustible of itself, will do little work if exposed to great heat; and in this is seen the great difference between our fire-proof buildings of the brick period and those of the iron period, and the inferior fire-proof qualities of the latter. The problem now is, to use the minimum of brick and the maximum of iron. I think, therefore, it must be conceded that with the best we can do with this material, there is danger; and the problem might be put thus: "Given Iron, make as nearly fire-proof buildings as possible out of it." What, then, has been done with it thus far? For columns, we have used cast tubes of all shapes and sizes and the wrought-iron pillars of the Phoenix Iron Company; for girders, we have used compound beams of cast-iron, with wrought ties—built up beams of various forms of rolled and plate iron, bolted and riveted together—and common rolled beams, used double; for floor beams we first used deck beams for wide spans and railroad iron for narrow spans; these have now been superseded by the I beam of various sizes. The Rolling Mills now have on their circulars I beams of great dimensions and suitable for girders, but refuse to fill any but large orders; indeed, I believe that only one mill has rollers for beams larger than thirteen inches, while the others will not put up machinery until they get large enough orders. So we are thus far deprived of large smooth beams of one piece, for girders of long span—beams which no one would desire to hide from view, but which might honestly tell their use to every beholder. For supports between beams we have had Peter Cooper's *terra cotta* pots and the four inch brick arches. The former are out of use and the latter are almost universally employed. Corrugated iron—first used in the Columbian Insurance building by Mr. Diaper—has also gone out of use. The destruction of the Fulton Bank, a so-called fire-proof building, sealed its fate as far as floors are concerned. ^[B] We have also had the experiment of stone floors in the American Exchange Bank, by Mr. Eidlitz, and repeated by another architect in the Mutual Benefit Life Insurance Building, at Newark, N. J. The stone slabs, brick arches, and the Parisian floors—of plaster or concrete, bedded upon bar iron gratings inserted between the beams—are the only practical systems of fire-proof floor construction, now

in use. The only attempt to lay the floor *on* the beams, of which I have knowledge, is in the sugar house above mentioned. This has suggested to me several methods of laying rigid floors upon beams at considerable spaces (three to five feet) from one another. Preliminary to so doing, I have above suggested the revival of the deck beam, or the I beam with a better form for the bottom flange, and the adoption of cast-iron shoes for the bearings.

The objections to the brick arches are that their great weight requires heavier beams than would otherwise be used, and that the form of their soffits is not beautiful; for they have the appearance of a long succession of little wagon vaults, requiring a resort to the doubtful expedient of furring the ceiling with iron lath. I think it might be objected to the French system of floors, that the expense would be too great, plaster being a dear article with us in comparison with its price in France, while our own cement has not the requisite properties to enable it to be substituted, besides being almost equally costly. The stone slabs, of Mr. Eidlitz, are the only rigid material thus far used successfully with iron beams, and could be used to better advantage if laid *on* the beams rather than resting upon their lower flanges, as is done in the American Exchange Bank. They are doubtless the handsomest material that can be used for this purpose, but are open to the objection of being heavy and expensive—where expense is a question, and utility only is sought—requiring heavy beams and calling for elaborate cutting on the under side. It will be pertinent to our inquiry, therefore, to ask if there are any other rigid materials adaptable to this purpose, and possessing the desired quality of lightness and cheapness. A former draughtsman of mine, now a member of the Institute, first suggested the use of slabs of slate, about two inches in thickness, for spans of four feet, and thicker or thinner in proportion to the distance of the beams from centres. I give his suggestion for what it is worth. But it led me to believe that we would eventually come to cast-iron as the practicable material for this purpose, possessing the requisite qualities of lightness and cheapness and capable of being bolted to the beams, thus answering all the purposes of flooring and bridging. Cast-iron plates may be used for flooring in two ways; first, when deafening and finished floor covering are required; second, when neither is required, as in manufacturing buildings, wherein a reasonably smooth flooring is required, and a few planks, laid where workmen habitually stand, will answer the purpose of non-conductors of

heat. Experiment must determine the minimum quantity of iron (in proportion to the strength required) to be used in the floor plates. In obtaining the proper form for strength, and to ensure true castings, the bottoms of the plates will naturally be covered with raised flanges, except at the edges, where they bear on the beams. These flanges or ribs may assume a decorative form, either a plain diaper or a larger pattern to form a complete design for the ceiling when many of them are combined. By a judicious arrangement of the flanges the actual thickness of the iron may be reduced to three-eighths, or a quarter of an inch. When deafening is required, strengthening flanges may also be cast on top of the plates, and consequently the beams can be placed at wide intervals. The flanges on the top will then serve to keep the concrete, used for deafening, in its place, and avoid the cracks which might occur in a large surface of cement. The deafening may be of any thickness required, and will serve as a bed for the floor tiles. All that is then required for the underside is judicious decoration of the beams and floor plates. When deafening is not required, as in manufacturing buildings, the tops should be smooth. It has been objected by a manufacturer, to whom I explained this system of construction, that the floors of iron would be too cold for the feet of workmen. But it would be very easy to put down platforms of wood where the men habitually stand. Besides, when the lower story is heated, the stratum of hot air immediately under the ceiling would naturally keep the floor at a higher temperature than that of the air in the room, and the greater conductivity of the iron would rather tend to warm the feet of those who stand upon it. The plates, in all cases, being bolted to the flanges of the beams, would serve as bridging for the floors.

By the above-described construction of floors, I would attempt to get rid of the obnoxious and expensive iron lath, so generally used. But it is more difficult to avoid their use on side walls, when the walls are to be plastered—and let me say here, that there can be no excuse for plastering the side walls in a fire-proof building, except for economy's sake. The easiest and by all means the cheapest expedient when plastering is required is to build four inch walls, secured to the main exterior walls by iron straps. These will not conflict with the building laws, provided you build your walls thick enough at the outset. There is, however, no better way in which to finish interior walls than to line them with stone or marble, or both combined. Where decorative effect is desired, I would use stone with marble panels.

Our native quarries now afford stone light enough in color to set at rest all objections that may be made to its use on the score of light. But if those should hold good the material might be marble paneled with marble, the former white, and the latter colored. Obviously the cheapest material for wall covering in natural materials would be slabs of white marble. Let us then make some comparison of figures, and see what can be done with this material. Iron lath, of the form generally used, cost \$1.25 per foot. Three coat plastering costs nine cents per foot. A responsible dealer in marble informs me that he will put up inch slabs of Italian veined or Vermont marble for one dollar and a half per foot. Which, then, would you choose, polished marble at \$1.50, or plaster, as good in appearance as that in any tenement house, at \$1.34? This is a fair comparison for exterior walls or ceilings. Italian marble slabs can be procured in any quantity, from eight to nine feet long and three feet wide. In a room fifteen feet high, allowing four feet for wainscot and two feet for cornice, you may line your walls with one length of marble.

What treatment do we now give to doors? We build brick jambs with wooden or iron lintels, as if we would treat the doors with wood. We then put up cast-iron jambs, rivet to their edges pilasters or architraves of the same material, and then surmount the whole perhaps, with a cast-iron cornice and pediment. Some have gone so far as to inlay the panels of the iron work with bits of colored marble, thus heightening the effect of the already rough finish of the iron, a roughness which the best foundrymen have been unable to prevent, and which, it would cost untold money to reduce down to the smoothness of ordinary work in pine wood. In one of our most pretentious houses on Fifth Avenue, they are now putting up jambs, architraves and cornices made of sawn slabs of marble or marble boards, in the same manner in which wood and iron have been used. And what does all this amount to? In the category of shams, there is no equal to this monstrous succession. You have imitated a Greek or Roman architrave and cornice by a wooden sham, your wooden sham has been imitated by an iron sham, your iron sham has been imitated by a marble sham; and what is the result? You have kept the form all along; you have come back to the original material by a succession of imitations, and have at last a shell without meat, marble carpentry instead of marble architecture. In all the stages of your attempt to revive the old forms, you have sham imitation of shams down to the final achievement of your carpenter in marble. Next

must follow, I suppose, the imitation marble-vender, who will crown the whole fabric of shams and give you something which can as much be called architecture as Mr. Shoddy's painted "red backs" and "blue backs" resemble standard literature. I offer no original suggestion to remedy this condition of affairs. Go back to your old Greek, go back to your old Roman models, if you like them, and seeing how they are built, go and do likewise; but spare us these sham contrivances. Set up your door posts and plant your lintel upon them, whether for exterior or interior use, and carve them to suit your fancy. They will be at least *good* so long as they be genuine and strong. Then figure up the cost of this kind of work, and see how much you have saved for your clients.

In conclusion, let me urge you to study diligently the various problems affecting this subject, which, in your experience, are continually offered for solution. In so doing, look mainly to a practical solution of the questions which may arise, and free yourselves from all consideration of so-called rules of art, which might control you. The development of architectural design was no less affected by local and circumstantial conditions, with the ancients, than it is with us; but the conditions at the present time are essentially different from, and decidedly more various than those which controlled our ancestors, whether of the classic or mediæval period. Whatever may have been achieved by art in those times, was the result of, and co-ordinate with the practical solution of problems then offered.

We have ignored the conditions which specially affect us, and the result is that our architecture, for whatever purpose, is without originality, and wholly irrational. As long as we allow ourselves to be governed by rules of art founded on the experience of the past, and precedents established by conditions which now do not exist, we need hope neither for good construction nor good art. The attempt to engraft the traditions of the past upon the practical work of this century has resulted in failures involving the waste of hundreds of millions of capital in this country alone; I might name from memory a score of buildings, many of them the most prominent, and all the most costly that have been erected, in proof of this assertion. I would commence with our national Capitol, in whose dome may be seen the most flagrant attempt in all modern time to perpetuate a traditionary style in a material entirely different from that in which the style was developed; so different that the foundations under it could not carry the superstructure, if

it were erected of the material for which it would appear to have been designed; and for want of foundations of sufficient breadth, even to carry the iron work, it has been necessary to carry the whole exterior iron colonnade upon iron brackets, concealed beneath what appears to be the podium for the whole dome, but which is in reality a box of thin plates of cast-iron, secured to a light framework, built out over the roof of the building.

In erecting modern fire-proof buildings, especially in so far as iron work is concerned, all the conditions imposed upon the architect are different from those which existed in past ages. The same may be said of the use of iron in any building. Subserviency to style, when the material used is not such as was the controlling element of that style, is destructive to all good art; for there can be no truly artistic effect except that which is produced by the best use of material, and its decoration in best accordance with its nature. If the use of iron is ever to lead to the erection of buildings worthy of being called works of art, such a result must be attained only by the recognition of this principle.

The best thinkers have doubted whether there can be any such thing as architecture in iron, assuming, of course, that to be called architecture, the material must be constructively used; and there is good reason for these doubts. An iron building does not always require the force of gravity to maintain the cohesion of its parts; it possesses such properties that it may be swung in the air or balanced on a single point, if it is necessary so to do. It is a machine admitting of as little decoration as a steam engine or a printing press. If iron alone were used for buildings, constructive necessity and economy combined, might lead us to build houses like steam boilers or water tanks.

What has been done thus far toward the erection of iron buildings on constructive principles? We can only recur to the buildings of the Crystal Palace pattern. We had a beautiful one in New York, admirably constructed, and well designed for its purpose; but even that building was decorated in the Moresque style, perhaps as nearly appropriate to the material employed as any that could have been selected. Here originality in treatment failed, just where it was wanted. The same constructive principles were involved in the design of this building which would have been involved in the erection of a fire-proof building. In this respect it was a success.

In the erection of fire-proof buildings, we are forced to do the best we can with iron while using it in the most varied capacities; but when its use can be spared, let me entreat you to rid yourselves of it; where it must be employed, use it rationally and constructively; but better not decorate it at all, than imitate styles not in harmony with its constructive properties. As all iron must be painted, I am inclined to believe that the best method of decorating it is in colors; for this treatment the iron must be plain and simple, and the colors may be proportionately brilliant. With regard to other materials, I would suggest nothing more than is said above—in all things build rationally. First, let your work be strong and well balanced—no part too heavy—no part too light. Then decorate it in harmony with its constructive features, never concealing materials, except where necessary to protect them, and emphasizing the main lines of the construction by ornamentation. Thus only can the great problem of the day be solved, and the fire-proof architecture of the nineteenth century be made worthy of a rational and progressive age.

NOTE.—An inspection of HARPER & BROTHERS' building, since writing this paper, has convinced me that the principle of division into horizontal compartments has been carried out more thoroughly in it than in any other building of the kind. There are no openings through the floors. It contains neither interior stairs nor hoistway; both are on the exterior. The stairs are in an isolated tower approached by bridges, and the hoistway is without enclosure. This arrangement is however extremely inconvenient.

FOOTNOTES:

[A] But by combustible material, I do not by any means intend what the insurance companies call hazardous, but dry goods, books, and similar things, which will burn independently of the building in which they are contained.

[B] (That disaster was owing also to the fact that the beams, other than girders, were made only of No. 12 sheet iron with flanges of 2 inch angle iron).

THE NEW MERCANTILE LIBRARY, PHILADELPHIA.

The history of the origin of public libraries is simple. Very few persons who possessed a desire to own books of great value could, in early times, afford to gratify their wish, owing either to want of the necessary means, or the very great scarcity of many works of intrinsic value. Before the invention of the great art of printing it is well known that all communicated learning was, of necessity, confined to manuscript on vellum. And that the only mode of repeating books was by transcription. The number of copies being extremely limited, it became necessary to have public places at each of which a copy might be placed for the use of those who desired to read, and as that number was in those days limited also, it was customary for some man of learning to read aloud to an audience.

These folios of manuscript, in time, accumulated to thousands, and the places of their deposit became institutions, and received the name of *librarium*. The term “librarian,” however, was applied in those days to the transcriber of books (*librarius*), rather than to the custodian, the latter officer being entitled *custos librariarum*, and who was nothing more than a janitor.

The enormous impetus given to education by the invention of printing, although it multiplied copies of books to such an extent as to render them cheap enough to become the property of individuals, still public libraries suffered no diminution, and the very increase of the draught seemed to promote the thirst for information, especially in that class in whom a taste for reading was controlled by a limit of means to become possessed of the necessary books. And although in our day the newspaper, the journal, and the serial, do much to disseminate knowledge among the millions, yet are libraries as much an institution of positive necessity as ever; for, in fact they whet the appetite for reading, and the brief paragraphs and condensed essays editorial are but so many stimulants to more extensive acquisitions of information. The taste grows, and the patronage of libraries increases, and such a progress must continue and enlarge whilst the mind of man lives

to accomplish the task set by Him in whose likeness the favored being is made.

The history of libraries is one of great interest to the lover of mental progress and the active civilization of our race, and might well call out the most industrious efforts of learned writers to do it justice. However, our business just now is with a local event—the inauguration of a new building by a most popular institution, the MERCANTILE LIBRARY of this city, which took place on the 15th of the past month, in the presence of a large and intellectual number of visitors of both sexes.

The rise and progress of this admirable institution is interesting. Started in 1822 in a small second-story room, with few books and fewer members, pinched to pay the rent of \$150 per annum, by degrees it gained vigor and steadily advanced to its present position, occupying now a building admirable in all the arrangements of room, light, heat, and ventilation.

This spacious building, occupying a prominent position on Tenth street, north of Chestnut, in this city, was purchased and fitted up at a cost of a quarter of a million dollars, and possesses a choice collection of books amounting to fifty-two thousand volumes, besides a well supplied news room, where will be found a great variety of journals from all parts of the civilized world, together with magazines, reviews, quarterlies, and annuals in abundance. The ladies having a separate department to themselves, unapproachable by the masculines.

The arrangement by which the reading rooms have been studiously kept in the rear of the building out of the reach of street disturbances, is one which gives it a great advantage over the public libraries of most other cities.

There is a well furnished chess room for the lovers of that mental game, and conversation, waiting and other rooms requisite to perfectly complete a truly desirable city institution.

We understand that the membership exceeds fifty thousand, and judging from what has been done, there is no reason to doubt its ultimately doubling that number in so large a city as this.

ARTIFICIAL STONE.—At the recent meeting of the Polytechnic Association of the American Institute, Mr. Thomas Hodgson exhibited and explained two methods of manufacturing and moulding artificial stone ornaments, blocks, etc., for buildings. One of these is prepared by treating lime with a solution of four ounces of oxalic acid in a gallon of water, thus producing an oxalate of lime, which is mixed with from two to four times its weight of sand. In this condition the material is a moist, friable powder. It is then moulded to the required form in Plaster-of-Paris moulds, removed from the latter, and suffered to dry. It is then preferably placed in a bath of dilute oxalic acid, which causes it to harden throughout, after which it is ready for use. In making the other variety, the inventor treats the oxalate of lime with a solution of silicate of potash, thus bringing it to a semi-fluid condition, whereupon it is poured into moulds and suffered to indurate.

Dr. Van der Weyde said that the oxalate of lime, being one of the most insoluble substances known in chemistry, its employment in the fabrication of artificial stone was a lucky thought. The use of potash and soda compounds for such purposes had been extensively attempted with very poor results, but the oxalate of lime was free from objections which hold good against such compounds.—*Railroad and Mining Register*.

THE NEW TREASURY BUILDING, at Washington, D. C., is now completed. This addition or north wing of the Treasury building is 65 by 195 feet, and occupies the site of the old State Department. The entire Treasury building covers an area of 520 by 278 feet, that is 144,550 square feet, or three acres and a half, including two large courts. On the eastern side of the building is a colonnade of thirty pillars, extending 336 feet north and south. On each of the other sides is a portico, each shaft of the columns of which is a monolith or single block of stone, 32 feet in height, and 4 feet 6 inches in diameter, that is 14 feet in circumference. The buttress caps, which partially inclose the steps of the porticoes, are single slabs of granite, 20 feet square by 2 feet thick. The granite was quarried on Dix's island, off the coast of Maine, and the larger slabs were taken to Washington in the rough, and there dressed. Fronting the north entrance is a fountain, the base of which is 12 feet in diameter, and the height 5 feet. It was cut from a single block of granite.

CORRESPONDENCE.

*It must be distinctly understood that we do not hold ourselves accountable
for the opinions of correspondents.*

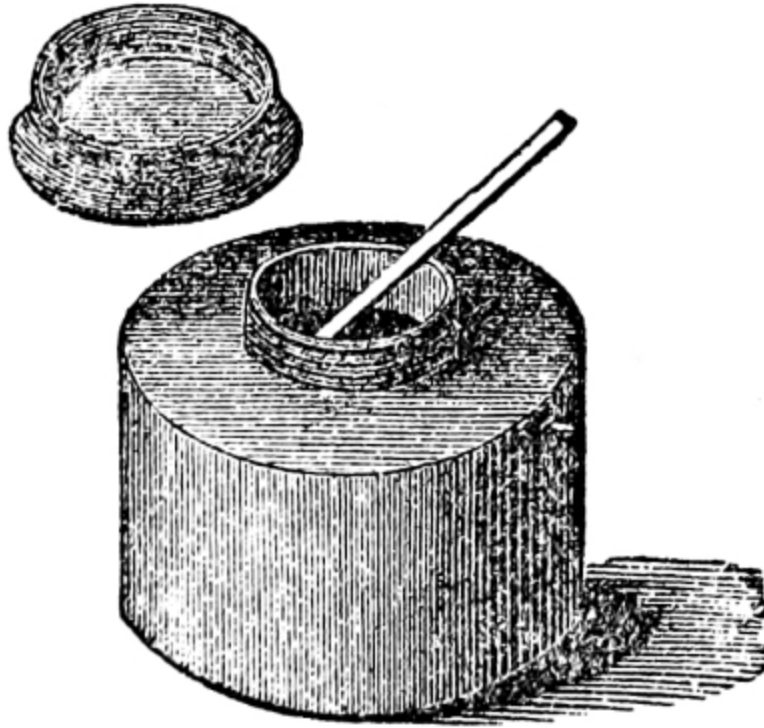
WASHINGTON,
July 20th, 1869.

“DEAR SIR:—Give your readers in your notes on Drawing and Drawing Materials, information that if a little powdered borax (borate of soda) is put into the water with which India Ink is rubbed up, and the mixture is kept in a tight bottle when not in use, it will keep sweet for months.

“The ink with which this is written was rubbed one year ago, and has sufficed for all my drawing during the past twelve months. A hard rubber ink bottle and screwed top has preserved, and it flows well, and the fragrance of the musk is as pleasant as when it was first rubbed.

“I have used the drawing pen for nearly forty years, and only a year ago was, by this receipt given me by a friend, relieved from the trouble of rubbing ink for every day’s work.

“Yours respectfully,
“M. C. MEIGS.”



We have assumed the liberty of giving the name of the writer of the foregoing excellent suggestions, in order to inspire learners with additional enthusiasm by showing them what an interest is taken in their progress by one who has attained to such a high position as the Quarter Master General of the United States Army, and we trust that Major General MEIGS' solicitude for art education may be emulated by many others, capable (if willing) of doing the cause an occasional service.

OMISSION.—In the preceding number of the REVIEW we overlooked the name of the architect who designed and superintended the *Atlantic* Hotel, of which we gave an illustrated description in the article on our "Sea Bathing Resorts." Unrequested by that gentleman, we think it but proper to give the credit to Mr. JOHN STEWART, Architect, of this city.

WEST PHILADELPHIA,
June 21st, 1869.

SIR:—It is to be hoped that at some time or other, before the appearance of cholera shall compel attention to the matter, we may see a move made in the direction of public baths. Is it not a little singular that our people of means who acknowledge the healthful necessity of bathing, and are lavish of expenditure to secure it in its utmost salt-water purity, do not seem to be aware of the advantage that blessing would be to hundreds of thousands of their fellow-beings, too poor to provide it for themselves. In winter we have Soup Associations, and other charitable societies. In summer can we not have Public Bathing Societies, just as necessary to the health of our community?

There are surely thousands who would subscribe their mite towards it; will not the millionaires lead off and set the ball in motion?

A CITIZEN.

Yes, we certainly think they ought, and we have not the slightest doubt but that they will, as soon as the coming man, who is to lead in this matter, shall make his appearance. Let us hope that person will soon be on hand.

Queries and Responses.

NEW YORK,
July 8th, 1869.

MR. EDITOR:—Is it not a most unaccountable fact that the New York Post Office structure, which was to have been commenced some two years since, is as seemingly a myth as one of those “castles in Spain,” of which we all have had at some time of our lives an idea. The site was duly purchased by the United States Government, designs were called for and provided in most eccentric profusion, and the select, if not elect, among the eighth-inch sealed suggestions were liberally paid for in awarded premiums. In fact all that is necessary to trumpet forth an advance was done. Where, then, is the new POST OFFICE? Out of sight—for, even the purchased site itself is not a certainty, then how should we expect to find the Post Office over ground when the ground is not yet decided on.

That Treasury Building at Washington is just now finished, after a lapse of time which makes gray hairs come on in unbidden numbers. Will the infant born this year, behold the promised New York Post Office before his growth of manhood is doubled into

“Lean and slipper’d Pantaloon?”

Alas, the “temporary” addition to that old Dutch church on Nassau street has but too truly proved a prediction, and we of the Empire city will either have to put up with the present arrangement, or build a postal structure of our own. It is evident now that the act of Congress, in this case made and provided, is but

A DEAD LETTER.

SARAH B.—In the case you mention, the lightning rod was secured to the wall of the house by iron staples. There was nothing to hinder the electric fluid from turning off on one of these, as it actually did. Accidents of a like

nature are constantly happening, and where sufficient precaution is not taken it would be far safer to have no conductor.

S. T., asks, is there any bank lock, of how many, and whatever combinations, that is absolutely secure against thoroughly posted and prepared burglars? We doubt that there is. For years the Bank of England trusted its vaults, filled with treasures, to the celebrated *Chubb* lock. Yet that ingenious Yankee, Hobbs, opened it in a surprising short time. The fact is—what man's ingenuity can make, man's ingenuity can also unmake.

C. G., Cincinnati.—We perfectly agree with you; the dwellings of this day are really combustible, and highly dangerous; much more so in fact than before burning fluids came so much into use. We also agree with you that the roofs of houses should not be of a material so liable to take fire on the occasion of a pyrotechnical display, or the passing of a spark-emitting locomotive.

Shingles could be easily rendered fire-proof by steeping them, before use, in a strong solution of alum. But most people would willingly “lose the sheep, to save the pennyworth of tar.”

R. D., Baltimore.—The silica coating of any building material renders it very durable. It is the combination of carbonate of lime, or chalk, with silicate of soda, or what is more commonly known as “soluble glass,” and by the old chemists called “oil of flint,” which, under heavy pressure, produces extraordinary hardness, and causes the great adherence of this cement to iron, brick, stone, or wood. And it is but one more proof of the practical property of the silicate, when applied to purposes such as those in which building most requires its valuable aid.

W. A., Ellsworth, Maine, asks for information as to the best manner of polishing instruments. We would recommend his getting a piece of buckskin and straining it on a square stick, covering one surface with

pulverized rotten-stone, or whiting, perfectly free from “grit.” For the instruments in which ink is used, having unscrewed and opened the hinged joints, clean off the ink first with a wet, then with a dry rag. Next rub the blades on the coated side of the buckskin, and lastly on the plain buckskin, until the appearance is satisfactory. We repeat that the pens should not be put away wet, but be carefully dried and rubbed on the buckskin after use. A drop of watchmakers’ oil on the screws and springs occasionally, will tend to insure the long and good service of instruments. Velvet is the best bed for them in the box; and the mould of their tray would be better cut out of cork than of wood. Any one can fit up his own instrument-case to suit his wants. Our advice is to buy only the instruments you have use for, *and get the best*, keeping them in constant order.

L., New York.—We agree with you, the names of streets should be painted on the lamps, and when a light of glass is broken and replaced the name should also be replaced.

S. R., Reading, Pa.—The idea is not new. Nay, it is as old as the hills. The ancients used hot air flues under their tiled floors. As long as we use boarded flooring we cannot do likewise, for reasons which any insurance office will freely give you.

PUBLICATIONS.

We have pleasure in acknowledging the receipt of the first number of THE ENGINEERING AND MINING JOURNAL, a weekly publication which was most desirable to our civil engineers in this country, who have hitherto had to depend for professional information on European sources. The American Journal of Mining was a popular periodical, and this prefixed addition to and modification of its title will go far to increase its well earned fame; for, judging by the specimen number, (and we know that cannot do its future full justice,) this new effort of Messrs. Western & Company is already a success as a most welcome co-laborer in the great constructive art. We tender it our best wishes, and place it on our exchange list.

MOORE'S RURAL NEW YORKER is an old and well tried friend of everything pertaining to agriculture and domestic economy. No country can boast of better serials of this class than ours, and foremost amongst the best we conscientiously place the Rural New Yorker. In its issue of July 10th, we find an illustrated suggestion for "a roomy house," in which we detect some defects which render its execution inadvisable. There is no provision for chimnies, and the stairs are impracticable. Such a house would be far more expensive than comfortable. However, it is pleasing to see men ready to contribute their mite to the general fund of information on a subject so intimately connected with home life and happiness.

THE SCIENTIFIC AMERICAN comes to us with its full share of the practical and the useful, amongst which we would particularly note an improved brick kiln. It has often surprised us to see the clumsy way in which bricks are usually burned and the serious waste of fuel arising from the loss of such a large percentage of heat, not to speak of want of uniformity in baking or burning. The kiln to which we allude is decidedly good and greatly superior to all its foreign predecessors, even Hoffman's, which it more nearly resembles.

HEARTH AND HOME, with all its attractiveness, is regularly on our table. This periodical is most creditable to the illustrated serial literature of our country, and we are satisfied of its being a fixed fact, from the evidence

before us of the liberality of its publishers and zeal of its gifted editors and staff of contributors. The prize song is a gem well worthy of a fitting setting in music equal to its own.

THE PRINTER'S CIRCULAR for July is filled with the interesting proceedings and intense enjoyment of the recent meeting of the National Union at Albany.

THE AMERICAN BUILDER for July has its usual amount of racy readings, its smart comments, and general information. It speaks well for the spirit of the western architects that our Chicago contemporary has laid in its foundation, and goes on with the work.

DESIGNS FOR STREET FRONTS, SUBURBAN HOUSES AND COTTAGES. By Cummings and Miller. This is a quarto volume containing fifty-two plates, with letter-press description of details for interior and exterior ornaments required in domestic architecture and the designs for the same. The former to a scale of a quarter inch, and the latter three-quarters of an inch to the foot. Besides this several designs are given for villas, country houses, and cottages. But the main advantage this work has over most of its predecessors, is in the very full and exhaustive hints, suggestions and instructions it gives to those in need of such; by which any practical man can readily apply any required embellishment to the house he proposes to construct. In fact the book before us supplies a very great want, by presenting to the builder remote from the professional aid of city architects an array of useful practical information which is inestimable to him, and is most desirable to the progress of tasteful construction throughout this wide country. The plates are unexceptionably executed, and the evident care with which this excellent guide to practical building has been put through the press renders it a most fitting work for those to whose wants it is so well adapted.

We highly recommend it as a faithful monitor and admirable assistant of the carpenter and builder. A. J. Bicknell, Troy, N. Y., is the publisher.

Transcriber's Notes:

The illustrations have been moved so that they do not break up paragraphs and so that they are next to the text they illustrate.

Typographical errors have been silently corrected but other variations in spelling and punctuation remain unaltered.

*** END OF THE PROJECT GUTENBERG EBOOK THE
ARCHITECTURAL REVIEW AND AMERICAN BUILDERS'
JOURNAL, AUG. 1869 ***

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